Chapter 2: Events, Samples, Probability, Bayes' Rule

Saturday, January 10, 2026 10:20 AM

Experiment : a process that generates data

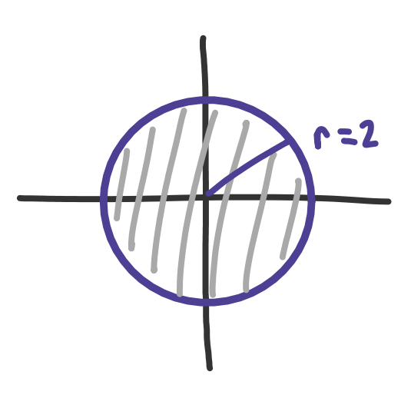
Sample space (S) : set of all potential outcomes of an experiment

e.g. select 3 items from production line, identify if they are defective (D) or not (N)

$S = \{DDD, DDN, DND, NDD, NND, NDN, DNN, NNN\}$

$2 \times 2 \times 2 = 2^3$ possibilities

e.g. $S = \{(x,y) \mid x^2 + y^2 \leq 4\}$



event : a subset of sample space (S)

$A = \{DDD, DDN, NDD, DND\}$

$B = \{NNN\}$

$C = \{(x,y) \mid x^2+y^2 = 4\}$

A "not"

A' (or A^c) : complement of A   ex. $A' = \{NNN, DNN, NDN, NND\}$

$A \cap B$: intersection of A & B (both occur)

$A \cup B$: union of A & B (at least one occurs)

$\emptyset$: null

If $A \cap B = \emptyset$, then A & B are **mutually exclusive** events

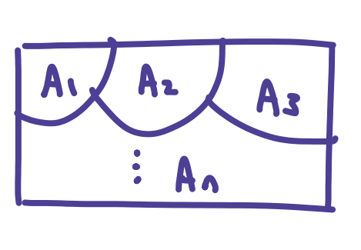
① $(A \cap B)' = A' \cup B'$
② $(A \cup B)' = A' \cap B'$
③ $A \cap \emptyset = \emptyset$
④ $A \cup \emptyset = A$

Probability $P(A)$: proportion of times the event A will occur in infinitely many repetitions of the experiment

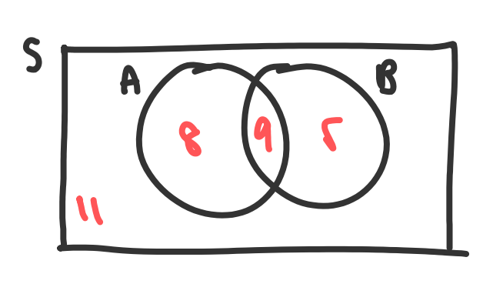
① $0 \leq P(A) \leq 1$
② $P(A) + P(A') = 1$
③ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
④ $P(A \cup B \cup C) = P(A) + P(B) + P(C)$
$\quad - P(A \cap B) - P(A \cap C) - P(B \cap C)$
$\quad + P(A \cap B \cap C)$

Defn: If $A_1, \dots A_n$ are mutually exclusive, then $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

In addition, if $A_1 \cup A_2 \cup \dots \cup A_n = S$, then $\{A_1, A_2, \dots A_n\}$ is a partition of S.



Ex. In a class of 33, there were 17 A's in the midterm, 14 in the final. 11 didn't earn A's on either. Calculate the prob that a randomly selected student earned A's on both exams.



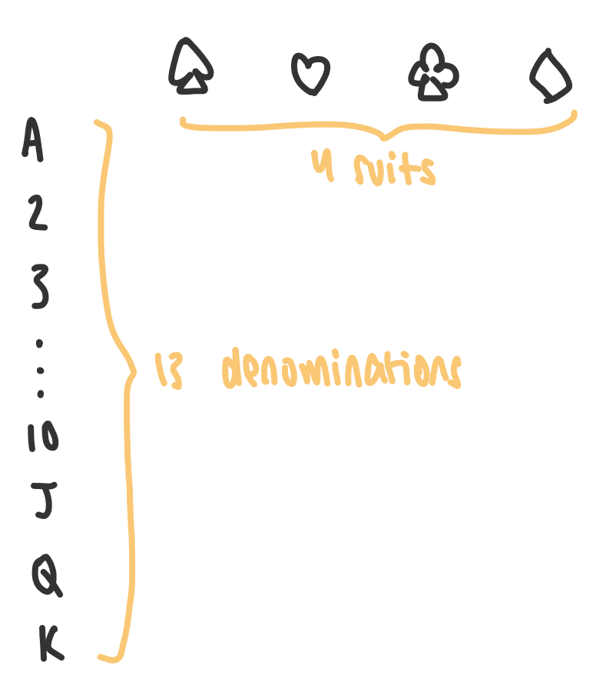
$17 + 14 + 11 - 33 = 9 \quad (A \cap B)$

$P(A \cap B) = \frac{9}{33}$

$\uparrow$ $\frac{\text{count of outcomes in } A \cap B}{\text{count of outcomes in } S}$

Equally likely outcomes : If there are k outcomes that are equally likely, then the prob of each is $\frac{1}{k}$

Ex. prob of 3 aces and 2 jacks in a poker hand



$$p = \frac{\binom{4}{3}\binom{4}{2}}{\binom{52}{5}}$$

Ex. P(any full house) = 3 of a kind & 2 of a kind

$$= \frac{\binom{13}{2}\binom{2}{1}\binom{4}{3}\binom{4}{2}}{\binom{52}{5}} \quad \text{or} \quad \frac{\binom{13}{1}\binom{4}{3}\binom{12}{1}\binom{4}{2}}{\binom{52}{5}}$$

Ex. P(4 of a kind) $= \dfrac{\binom{13}{1}\binom{4}{4}\overset{\text{fifth card}}{\binom{48}{1}}}{\binom{52}{5}}$

Ex. P(exactly one pair) $= \dfrac{\underbrace{\binom{13}{1}\binom{4}{2}}_{\text{one pair}}\underbrace{\binom{12}{3}\binom{4}{1}\binom{4}{1}\binom{4}{1}}_{\text{3 non-pairs}}}{\binom{52}{5}}$

when order matters : ${}_nP_r = \frac{n!}{(n-r)!}$ (permutation)

when order doesn't matter: $\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$ (combination)

Ex. pair of fair dice, prob that second > first?

6 / 5 4 3 2 1
5 / 4 3 2 1
4 / 3 2 1
3 / 2 1
2 / 1

$\frac{15}{6 \times 6} = \frac{15}{36}$

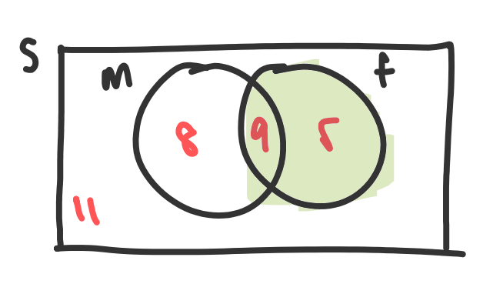
Ex. b boys and g girls lined up in random order. Prob i^{th} position is a girl?

$$\frac{g}{g+b} = \frac{\binom{g}{1}(b+g-1)!}{(b+g)!}$$

Conditional probability :

$P(B \mid A)$ = prob of event B occurring when it's known that event A occurred

Ex.



$P(m \mid f) = \frac{9}{14}$

Defn : $P(B \mid A) = \frac{P(A \cap B)}{P(A)}$, $P(A) > 0$

Ex. Answer to "do you drink"

	yes (D)	no (N)	total
male (M)	19	41	60
female (F)	12	28	40
total	31	69	100

probability of randomly selected individual being male who drinks :

$P(M \cap D) = \frac{19}{100}$

$P(M \mid D) = \frac{19}{60}$

$P(D \mid M) = \frac{19}{31}$

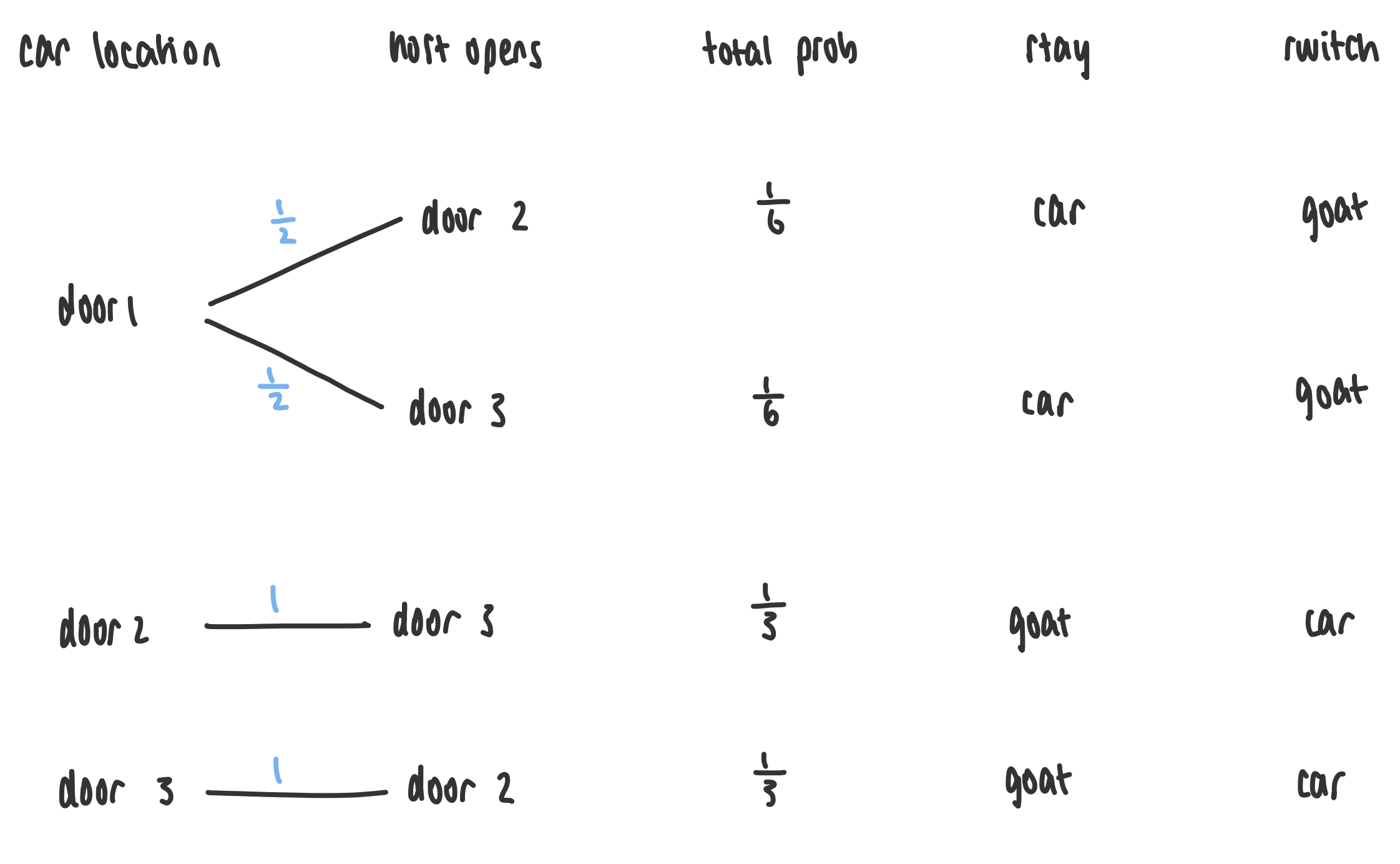
Thrm : If the events $B_1, \dots B_k$ constitute a partition of S such that $P(B_i) \neq 0$ for $i = 1, \dots k$, then for any event A of S, $P(A) = \sum_{i=1}^{k} P(B_i \cap A)$

since all B_i are mutually exclusive, so all $B_i \cap A$ are mutually exclusive

$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_k)$

$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_k)$

Ex. Monty Hall problem: you're on a game show, given 3 choices of doors. Behind one is a car, behind the others goats.
You pick a door (say #1), Monty opens another door, say #3.
Asks if you want to switch (to #2). What do you say?



car location	host opens	total prob	stay	switch
door 1	door 2	$\frac{1}{6}$	car	goat
	door 3	$\frac{1}{6}$	car	goat
door 2	door 3	$\frac{1}{3}$	goat	car
door 3	door 2	$\frac{1}{3}$	goat	car

$P(\text{car} \mid \text{switch}) = \frac{1}{3}$

Ex. Ignore leap year, assume all birthdays are equally likely. What's the prob that at least two people have the same birthday in n=23 people?

If $n = 366$, $p = 1$
⋮
$n = 57$, $p = 0.99$
⋮
$n = 23$, $p = 0.507$

$$P(\text{at least one match}) = 1 - P(\text{no match}) = 1 - \underbrace{\frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdots \frac{(366-n)}{365}}_{n-1}$$

Defn: Independence

A and B are independent $\Longleftrightarrow$ $P(B \mid A) = P(B)$
$P(A \mid B) = P(A)$
$P(A \cap B) = P(A)P(B)$

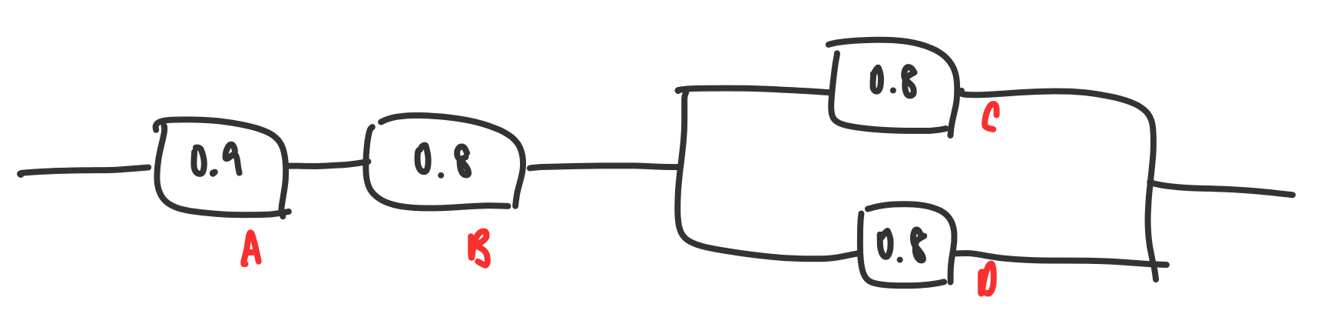
Defn: If events $A_1, A_2, \dots A_k$ are independent, then

$P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1)P(A_2)\dots P(A_k)$

Defn: A collection of events $A_1, \dots A_n$ are mutually independent if and only if for any subcollection $A_{i1}, \dots A_{ik}$, we have $P(\bigcap_{j=1}^{k} A_{ij}) = \prod_{j=1}^{k} P(A_{ij})$

e.g. $P(A_1 \cap A_2) = P(A_1)P(A_2)$
$P(A_1 \cap A_3) = P(A_1)P(A_3)$
$P(A_2 \cap A_3) = P(A_2)P(A_3)$
$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$
$\Longleftrightarrow$ A_1, A_2, A_3 are mutually independent

Ex.



$P(\text{system works}) = P(A)P(B)P(C)P(D) + P(A)P(B)P(C)(1-P(D)) + P(A)P(B)(1-P(C))P(D)$

$= (0.9)(0.9)(0.8)(0.8) + (0.9)(0.9)(0.8)(0.2) + (0.9)(0.9)(0.2)(0.8)$

$= 0.7776$

$P(C \text{ doesn't work} \mid \text{system works}) = \frac{P(C \text{ doesn't work} \cap \text{system works})}{P(\text{system works})}$

$= \frac{P(A)P(B)(1-P(C))P(D)}{P(\text{system works})}$

$= \frac{0.9 \cdot 0.9 \cdot 0.2 \cdot 0.8}{0.7776}$

$= 0.1\overline{6}$

Theorem of total probability :

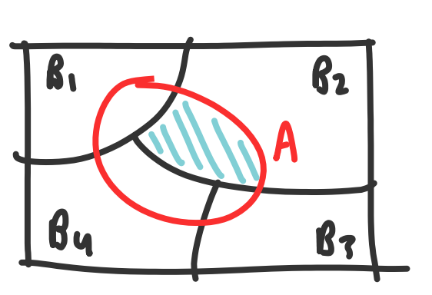
If events $B_1, \dots B_k$ is a partition of S such that $P(B_i) \neq 0$ for $i = 1, \dots k$ then for any event A of S,

$$P(A) = \sum_{i=1}^{k} P(B_i \cap A) = \underbrace{\sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)}_{\text{based on cond. prob. defn.}}$$

Theorem: Bayes Rule

If events $B_1, \dots B_k$ constitute a partition of S such that $P(B_i) \neq 0$ for all $i = 1, \dots k$ then for any event A in S such that $P(A) \neq 0$,

$$P(B_r \mid A) = \frac{P(B_r \cap A)}{\sum_{i=1}^{k} P(B_i \cap A)} = \frac{P(B_r)\, P(A \mid B_r)}{\sum_{i=1}^{k} P(B_i) P(A \mid B_i)} \quad \text{for } r = 1, \dots k$$



Chapter 3: Random Variables, Distributions, Statistical Independence

Monday, January 19, 2026 11:09 AM

Defn: A random variable (r.v.) is a function that associates a real number with each element in S.

Ex. Three electronic component are tested.

$$S = \left\{ \begin{matrix} NNN, & DNN, & DDN, & DDD, \\ NND, & NDD, & NDN, & DND \end{matrix} \right\}$$

X : # of defectives (a random variable)

$X=0$: $\{NNN\}$

$X=1$: $\{NND, NDN, DNN\}$

$X=2$: $\{DDN, DND, NDD\}$

$X=3$: $\{DDD\}$

Ex. One component is tested. $S = \{D, N\}$

$$X = \begin{cases} 1, & \text{if component } D \\ 0, & N \end{cases}$$ → dummy variable, outcome categorical in nature

Defn: A r.v. is called discrete if its set of possible outcomes is countable (finite or infinite)

Ex. Sample components until observe a D.

$S = \{D, ND, NND, NNND, \dots\}$

X : # of samples tested until we observe the first D.

$X=1$: $\{D\}$

$X=2$: $\{ND\}$

$\vdots$ $X = 1, 2, 3, \dots$

Defn: $f(x)$: probability mass function (pmf) of the discrete r.v. X such that for each possible outcome $X = x$,

① $P(X = x) = f(x)$

② $f(x) \geq 0$, $\sum_x f(x) = 1$

X : uppercase (concept of r.v.)
x : lowercase (instance of r.v.)

Ex. one component:

x	0	1
$P(X=x)$	$1-p$	p ← prob. of observing D

Bernoulli random variable

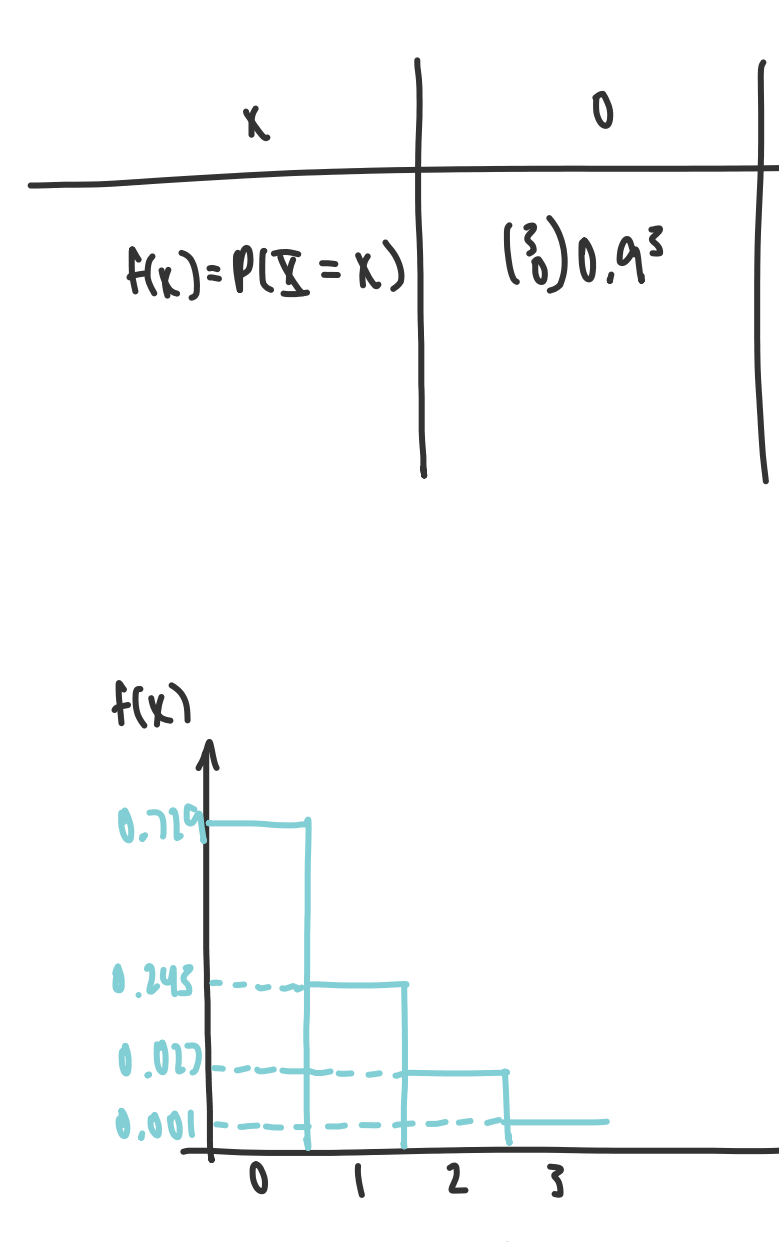
Ex. assume $p=0.1$ and 3 components tested independently

X = # of defectives

x	0	1	2	3
$f(x)=P(X=x)$	$\binom{3}{0}0.9^3$	$\binom{3}{1}(0.1)(0.9)^2$	$\binom{3}{2}(0.1)^2(0.9)$	$\binom{3}{3}(0.1)^3$

← should add up to 1

Binomial random variable



probability histogram

Ex. sample components until reach a D. ($p=0.1$)

X : number of samples collected

x	$P(X=x)=f(x)$
1	0.1
2	$(0.9)(0.1)$
3	$(0.9)^2(0.1)$
$\vdots$	$\vdots$
x	$(0.9)^{x-1}(0.1)$
$\vdots$	$\vdots$

$$\sum_x 0.9^{x-1} \cdot 0.1 = 0.1(1 + 0.9 + 0.9^2 + \dots) = 0.1\left(\frac{1}{1-0.9}\right) = 1$$

geometric random variable

Defn: The culminative distribution function (cdf) of a discrete function X with prob $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t) \quad \text{for } -\infty < x < \infty$$

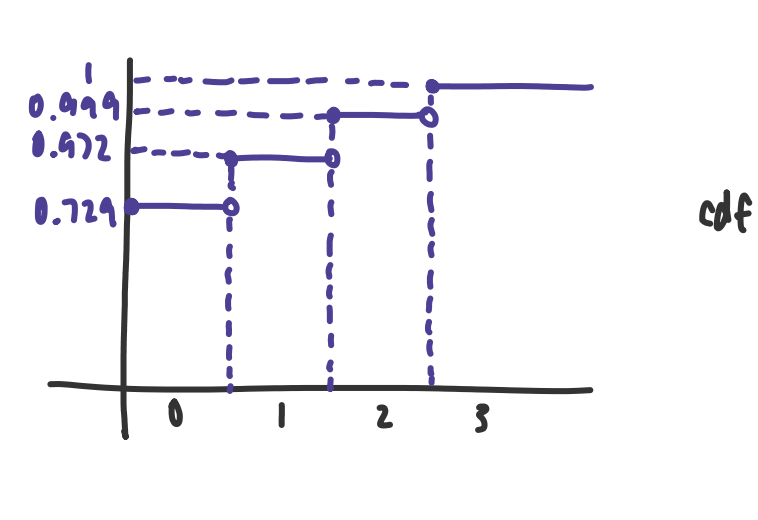
Ex. Three components tested ($p = 0.1$)

$F(0) = P(X \leq 0) = 0.729$

$F(1) = P(X \leq 1) = f(0) + f(1) = 0.729 + 0.243$

$F(2) = P(X \leq 2) = f(0) + f(1) + f(2)$

$F(3) = P(X \leq 3) = 1$



cdf

Note: cdf is a monotone non-decreasing function → $F(x) \leq F(y)$ if $x < y$

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.729, & 0 \leq x < 1 \\ 0.972, & 1 \leq x < 2 \\ 0.999, & 2 \leq x < 3 \\ 1, & 3 \leq x \end{cases}$$

for $a < b$, $\boxed{P(a < X \leq b) = F(b) - F(a)}$

$P(0 < X \leq 2) = P(X=1) + P(X=2) = 0.243 + 0.027 = 0.27$

$P(0 < X \leq 2) = F(2) - F(0) = 0.999 - 0.729 = 0.27$

If S contains an infinite number of possible outcomes equal to the number of points of a line segment, it is called a continuous sample space.

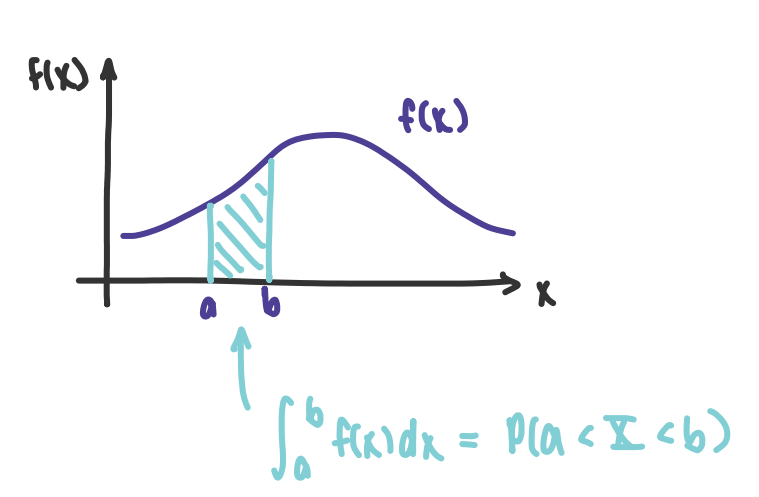
A continuous r.v. has a probability of zero of assuming exactly any of its values. So we look at intervals.

Defn: The function $f(x)$ is a probability density function (pdf) of the continuous r.v. X defined over the set of real numbers $\mathbb{R}$ if:

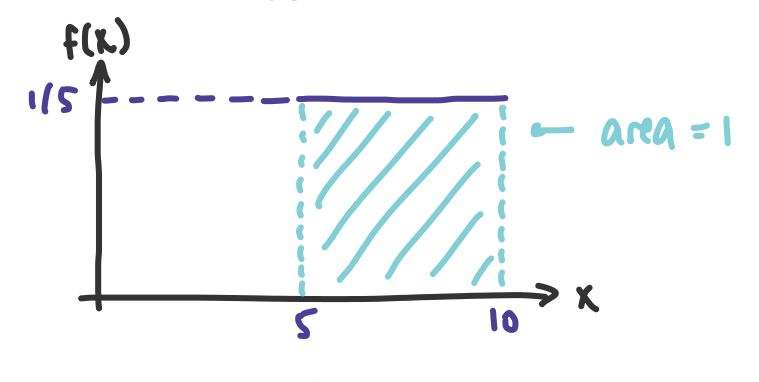
(1) $f(x) \geq 0$ for all $x \in \mathbb{R}$

(2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(3) $P(a < X < b) = \int_a^b f(x)\,dx = \underbrace{P(a \leq X \leq b) = P(a < x \leq b) = P(a \leq x < b)}_{\text{doesn't matter if inclusive/exclusive}}$



$\int_a^b f(x)dx = P(a < X < b)$



$$f(x) = \begin{cases} 0, & \text{elsewhere} \\ \frac{1}{5}, & 5 < x < 10 \end{cases}$$

$$P(X < 7) = \int_5^7 \frac{1}{5}\,dx = \frac{2}{5}$$

Defn: Cumulative density function (CDF)

The function $F(x)$ of a continuous r.v. X with pdf $f(x)$ if

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt \quad \text{for } -\infty < x < \infty$$

consequently,

$$f(x) = \frac{dF(x)}{dx}, \quad P(a < X < b) = F(b) - F(a)$$

Ex. X: time until chemical reaction is completed (in msec)

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-0.01x}, & 0 \leq x \end{cases}$$

(a) find pdf:

$$\frac{dF(x)}{dx} \qquad f(x) = \begin{cases} 0, & x < 0 \\ 0.01e^{-0.01x}, & 0 \leq x \end{cases}$$

(b) $P(X < 200) = F(200) = 1 - e^{-2} = 0.8647$

(c) is this a valid cdf?

as $x \to -\infty$, $F(x) \to 0$

as $x \to +\infty$, $F(x) \to 1$

monotone non-decreasing

$F(x) \leq F(y)$ if $x \leq y$

Joint probability distributions

Defn: The function $f(x,y)$ is a joint pmf of discrete r.v.'s X and Y if

(1) $f(x,y) \geq 0$ for all (x,y)

(2) $\sum_x \sum_y f(x,y) = 1$

(3) $P(X=x, Y=y) = f(x,y)$

For any region A in the xy plane, $P[(X,Y) \in A] = \sum_A \sum f(x,y)$

Ex. Two refills for a pen are selected at random from a box that contains 3 blue, 2 red, and 3 green refills.

X : # of blue refills selected

Y : # of red refills selected

	$f(x,y)$	x = 0	1	2	row totals
	0	$\frac{\binom{3}{2}}{\binom{8}{2}} = \frac{3}{28}$	$\frac{\binom{3}{1}\binom{3}{1}}{\binom{8}{2}} = \frac{9}{28}$	$\frac{\binom{3}{2}}{\binom{8}{2}} = \frac{3}{28}$	15/28
y	1	$\frac{\binom{2}{1}\binom{3}{1}}{\binom{8}{2}} = \frac{3}{14}$	$\frac{3}{14}$	0	5/7
	2	$\frac{\binom{2}{2}}{\binom{8}{2}} = \frac{1}{28}$	0	0	1/28
	column totals	5/14	15/28	3/28	1

marginal pmfs

marginal pmfs

Defn: marginal pmf of X is

$$g(x) = \sum_y f(x,y)$$

for Y,

$$h(y) = \sum_x f(x,y)$$

$f_{X,Y}(x,y)$ f_X f_Y (notation)

① $P(Y=1 \mid X=1) = \frac{P(X=1, Y=1)}{P(X=1)} = \frac{3/14}{15/28}$

② $P(Y=0 \mid X=1) = \frac{9/28}{15/28}$

③ $P(Y=2 \mid X=1) = 0$

① + ② + ③ = 1

Defn: The conditional distribution of Y given $X = x$ is

$$f(y \mid x) = \frac{f(x,y)}{g(x)}, \quad g(x) > 0$$

$$f(x \mid y) = \frac{f(x,y)}{h(y)}, \quad h(y) > 0$$

holds for both discrete and cont. r.v.'s.

Defn: The function $f(x,y)$ is a joint pdf of continuous r.v.'s X and Y if

(1) $f(x,y) \geq 0$ for all (x,y)

(2) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

(3) $P[(X,Y) \in A] = \iint_A f(x,y)\,dx\,dy$ for any region A in xy plane

Joint pdf is a surface lying above the xy plane.
Probability is the volume under the surface

Ch. 3 Review

1) r.v. X

2) pmf $f(x) = P(X = x)$, discrete

3) pdf $\int_a^b f(x)dx = P(a < X < b)$

4) cdf $F(x) = P(X \leq x)$

5) joint distributions $f(x,y)$
marginal distributions

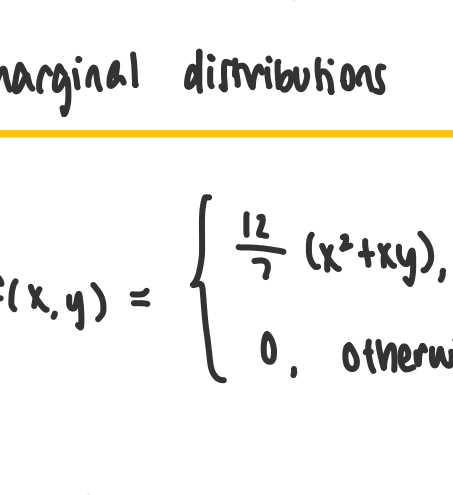
Ex.

$$f(x,y) = \begin{cases} \frac{12}{7}(x^2+xy), & 0 \leq x \leq 1,\ 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

a) $\int_0^1 \int_0^1 \frac{12}{7}(x^2+xy)\,dx\,dy = 1$

b) $P(0 < X < 0.2,\ 0.3 < Y < 1) = \int_{0.3}^{1} \int_0^{0.2} \frac{12}{7}(x^2+xy)\,dx\,dy$

c) $P(X > Y) = \int_0^1 \int_0^x \frac{12}{7}(x^2+xy)\,dy\,dx$



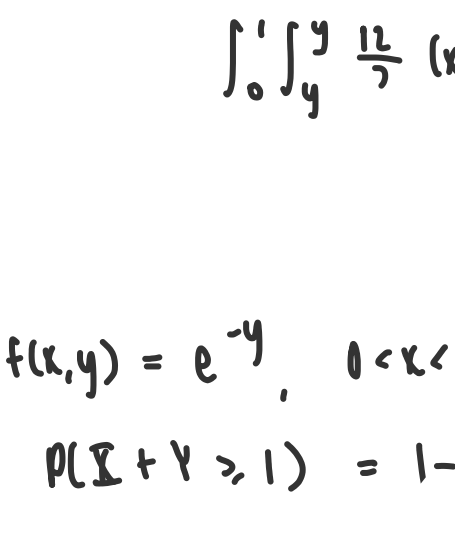
d) $P(X = Y) = 0$

$\int_0^1 \int_y^y \frac{12}{7}(x^2+xy)\,dx\,dy = 0$

Ex. $f(x,y) = e^{-y}$, $0 < x < y < \infty$

$P(X + Y \geq 1) = 1 - P(X + Y < 1) = \int_0^{1/2} \int_x^{1-x} e^{-y}\,dy\,dx$

$= 2e^{-1/2} - e^{-1}$



Defn. Marginal pdf's of X and Y are:

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Ex. continued from previous

e) $g(x) = \int_0^1 \frac{12}{7}(x^2+xy)\,dy = \frac{12}{7}\left(x^2 + \frac{x}{2}\right)$, $0 \leq x \leq 1$

$h(y) = \int_0^1 \frac{12}{7}(x^2+xy)\,dx$

f) $f(\overset{\text{y is r.v.}}{y} \mid \underset{\text{x is known, not r.v. it's a constant}}{x}) = \frac{f(x,y)}{g(x)} = \frac{\frac{12}{7}(x^2+xy)}{\frac{12}{7}(x^2+\frac{x}{2})}$ $\quad 0 \leq y \leq 1$, $0 < x \leq 1$

Defn. X and Y are said to be statistically independent $\iff f(x,y) = g(x)h(y)$ for all (x,y) within their range.

If X and Y are independent, then $f(x \mid y) = g(x)$

Ex. Recall refill example

		x = 0	1	2	
	0	3/28	9/28	3/28	15/28
y	1	3/14	3/14	0	3/7
	2	1/28	0	0	1/28
		5/14	15/28	3/28	1

$h(y)$

$g(x)$

$f(0,1) \neq g(0)h(1)$

$\frac{3}{14} \neq \frac{5}{14} \cdot \frac{3}{7}$ ⇒ not independent

Ex. X and Y are discrete r.v.'s whose values are nonnegative integers.

$$f(x,y) = \frac{1}{x!\,y!} \lambda^x \mu^y e^{-(\lambda+\mu)} \qquad x, y = 0, 1, 2, \dots$$

$f(x,y) \overset{?}{=} g(x)h(y)$

$$g(x) = \sum_{y=0}^{\infty} \frac{1}{x!\,y!} \lambda^x \mu^y e^{-(\lambda+\mu)} = \frac{1}{x!} \lambda^x e^{-(\lambda+\mu)} \underbrace{\sum_{y=0}^{\infty} \frac{\mu^y}{y!}}_{e^{\mu}} = \frac{1}{x!} \lambda^x e^{-\lambda}$$

$$h(y) = \sum_{x=0}^{\infty} f(x,y) = \frac{1}{y!} \mu^y e^{-\mu} \qquad y = 0, 1, \dots$$

Factorization of $f(x,y)$ to functions of x and y is enough for independence but the set of points where the point density (or pmf) is positive must be the Cartesian product of the set of points where marginals are positive.

$f(x,y)$ = function (x) * function (y) * constant =1?

Ex.

$$f(x,y) = \begin{cases} 4(x^2+y^2), & x, y > 0,\ x+y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

are X and Y independent? $f(x,y) \overset{?}{=} g(x)h(y)$

Defn: Let $X_1, \dots X_n$ be n random variables, discrete or cont., with joint probability distribution $f(x_1, \dots x_n)$ and marginals $f_1(x_1), f_2(x_2), \dots f_n(x_n)$

$X_1, \dots X_n$ are said to be mutually statistically independent $\iff f(x_1, \dots x_n) = f_1(x_1) f_2(x_2) \dots f_n(x_n)$

Ex. Shelf life of certain food product is r.v. X

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

X_1, X_2, X_3 represent shelf lives of 3 containers selected independently.

Find $P(X_1 < 2,\ 1 < X_2 < 3,\ X_3 > 2) = P(X_1 < 2)P(1 < X_2 < 3)P(X_3 > 2)$

$= \int_0^2 e^{-x}dx \cdot \int_1^3 e^{-x}dx \cdot \int_2^{\infty} e^{-x}dx$

$= 0.0372$

or $f(x_1, x_2, x_3) = f(x_1) f(x_2) f(x_3) = e^{-x_1} e^{-x_2} e^{-x_3}$, $x_1, x_2, x_3 > 0$

$P(X_1 < 2, \dots) = \int_0^2 \int_1^3 \int_2^{\infty} e^{-x_1} \cdot e^{-x_2} \cdot e^{-x_3}\,dx_1\,dx_2\,dx_3$

Chapter 4: Expectation, Variance, Covariance,

Thursday, January 29, 2026 10:50 AM

Ex. Roll a die repeatedly, what average value is expected?

In 6000 000 rolls, expect 1000 000 of each side.

$$\text{Average} = \frac{1000\,000\cdot 1 + 1000\,000\cdot 2 + \dots + 1000\,000\cdot 6}{6000\,000}$$

$$= 3.5$$

Defn. for a r.v. X, the expectation (or expected value or mean) of X is the expected average value of X in the long run

$$\mu = E(X) = \begin{cases} \sum_x x f(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} x f(x)\,dx, & X \text{ cont.} \end{cases}$$

Ex. number of messages per hour sent over a network has pmf

x	$f(x)$
10	0.08
11	0.15
12	0.3
13	0.2
14	0.2
15	0.07

$$\mu = E(X) = 10\times 0.08 + 14\times 0.15 + 12\times 0.3 + 13\times 0.2 + 14\times 0.2 + 15\times 0.07$$
$$= 12.5$$

Ex. Deal or no deal

$\boxed{\$1}$ or $\boxed{\$10000}$

fair offer from bank:

$$\text{Expected value} = 0.5\times 1 + 0.5\times 10000$$
$$= \$5000.5$$

Ex. X is r.v. that denotes life in hours of a certain electronic device

$$f(x) = \begin{cases} \frac{20000}{x^3}, & x > 100 \\ 0, & \text{elsewhere} \end{cases}$$

$$\mu = E(X) = \int_{100}^{\infty} x\,\frac{20000}{x^3}\,dx = 200$$

Device lasts on average 200 hours

Ex. X a discrete r.v. with pmf $f(x)$ $x = -1, 0, 1, 2$

$g(X)$ a new r.v. $g(X) = X^2$

x	-1	0	1	2
$f(x)$	$f(-1)$	$f(0)$	$f(1)$	$f(2)$

$g(x) = x^2$	0	1	4
$P[g(X) = g(x)]$	$f(0)$	$f(-1) + f(1)$	$f(2)$

$$E[g(X)] = 0\cdot f(0) + 1\cdot [f(-1) + f(1)] + 4\cdot f(2)$$
$$= 0^2\cdot f(0) + 1^2\cdot f(-1) + 1^2\cdot f(1) + 2^2\cdot f(2)$$
$$= \sum_x x^2 f(x) = \sum_x g(x) f(x)$$

Thrm: Let X be a r.v. with prob dist. $f(x)$

The expected value of the r.v. $g(X)$, ie. a function of X

$$\mu = E[g(x)] = \begin{cases} \sum_x g(x) f(x) & \text{if disc.} \\ \int_{-\infty}^{\infty} g(x) f(x)\,dx & \text{if cont.} \end{cases}$$

Ex. Bowl contain 5 chips

3 are marked \$1 each

2 \$4 each

If A costs \$4.75 to play the game, would you participate?

Game: A player draws 2 at random and is paid there

X : number of \$1 chips drawn $x = 0, 1, 2$

$$f(x) = \begin{cases} \frac{\binom{3}{x}\binom{2}{2-x}}{\binom{5}{2}}, & x = 0, 1, 2 \\ 0, & \text{elsewhere} \end{cases}$$

payout: $g(x) = x + 4(2-x)$
$= 8 - 3x$ dollar

$$E(g(X)) = E[8-3x] = \sum_{x=0}^{2} (8-3x) f(x) = \$4.40$$

Ex. $f(x) = 4x^3$, $0 < x < 1$

$$E(X) = \int_0^1 x\,4x^3\,dx$$
$$E(X^2) = \int_0^1 x^2\,4x^3\,dx$$

Ex. $f(x,y) = x(1+3y^2)$, $0 < x < 2$, $0 < y < 1$

$$E(X) = \int_0^2 x \underline{f(x)}\,dx$$

$f(x) = \int_0^1 f(x,y)\,dy$

$$= \int_0^2 x \left(\int_0^1 x(1+3y^2)\,dy\right) dx$$
$$= \int_0^2 \int_0^1 x^2(1+3y^2)\,dy\,dx$$
$$= \int_0^2 \int_0^1 x f(x,y)\,dy\,dx$$

$$E\left(\frac{Y}{X}\right) = \int_0^1 \int_0^2 \frac{y}{x} f(x,y)\,dx\,dy$$

Properties of expectation E

for X, Y r.v.'s and $a, b \in \mathbb{R}$ constants.

① $E(aX + b) = aE(X) + b$

cont case: $E(aX+b) = \int_{-\infty}^{\infty} (ax+b) f(x)\,dx = a\int x f(x)\,dx + b\int f(x)\,dx$

② $E(a) = a$

③ $E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)]$

proof: $\int_{-\infty}^{\infty} [g(x) \pm h(x)] f(x)\,dx = \int_{-\infty}^{\infty} g(x) f(x)\,dx + \int_{-\infty}^{\infty} h(x) f(x)\,dx$

④ If X is a non-negative r.v. then $E(X) \geq 0$

⑤ If X is non-negative r.v. and $E(X) = 0$ $\Leftrightarrow$ $P(X = 0) = 1$

Variance:

$E(X)$ is a measure of "center" of distribution.

Variance (σ^2), $\text{var}(X)$, is a measure of how closely concentrated the distribution is to the center (μ) (mean, not median/mode)

$\sigma^2 = E[(X-\mu)^2]$ where $\mu = E(X)$

$$= \begin{cases} \sum_x (x-\mu)^2 f(x), & \text{disc.} \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx, & \text{cont.} \end{cases}$$

σ: standard deviation

Thrm: $\text{var}(X) = \sigma^2 = E(X^2) - \mu^2$

proof:
$$E[(x-\mu)^2] = E[x^2 - 2x\mu + \mu^2]$$
$$= E[x^2] - E[2\mu x] + E[\mu^2]$$
$$= E(x^2) - 2\mu \underbrace{E[x]}_{\mu} + \mu^2$$
$$= E[x^2] - \mu^2$$

Ex. weekly demand for a drink, in thousand liters, from a store is a cont. r.v. X

$f(x) = 2(x-1)$, $1 < x < 2$

$$\mu = E[x] = \int_1^2 x f(x)\,dx$$
$$= \int_1^2 2x^2 - 2x\,dx$$
$$= \frac{5}{3}$$
$$E[X^2] = \int_1^2 x^2 f(x)\,dx$$
$$= \frac{17}{6}$$
$$\text{var}(X) = E[X^2] - \mu^2$$
$$= \left(\frac{17}{6}\right)^2 - \left(\frac{5}{3}\right)^2$$
$$= \frac{1}{18}$$

Thrm: X a r.v. with dist. $f(x)$

$g(X)$, a function of X, another r.v.

$$\underbrace{\sigma^2_{g(X)}}_{\text{var}(g(X))} = E\{[g(X) - \mu_{g(X)}]^2\}$$
$$= E[g^2(X)] - \underbrace{E^2[g(X)]}_{\text{"}\mu_{g(X)}^2\text{"}}$$

$$E[(\underset{g(X)}{X} - \underset{E[g(X)]}{\mu})^2] = \text{var}(\underset{g(X)}{X})$$

Properties of variance

X, Y r.v's $a, b \in \mathbb{R}$ constants

① $\text{var}(aX + b) = a^2\,\text{var}(X)$

proof:
$$E[(aX+b)^2] - [E(aX+b)]^2$$
$$= E[a^2X^2 + 2abX + b^2] - [aE[X] + b]^2$$
$$= a^2E[X^2] + \cancel{2abE[X]} + \cancel{b^2} - a^2E^2[X] - \cancel{2abE[X]} - \cancel{b^2}$$
$$= a^2\underbrace{[E[X^2] - E^2[X]]}_{\text{var}(X)}$$
$$= a^2\,\text{var}(X)$$

② $\text{var}(X) \geq 0$

③ $\text{var}(a) = 0$

④ $\text{var}(X) = 0$ $\Leftrightarrow$ $X = E[X]$ with prob 1 (it is a constant)

Thrm: X, Y r.v.'s with joint dist. $f(x,y)$

$\left.\begin{matrix} g(X,Y) \\ h(X,Y) \end{matrix}\right\}$ functions of X, Y

$$E[g(X,Y) \pm h(X,Y)] = E[g(X,Y)] \pm E[h(X,Y)]$$

⑤ $\text{var}(aX + bY) = a^2\,\text{var}(X) + b^2\,\text{var}(Y) + 2ab\,\underbrace{E[(X - E(X))(Y - E(Y))]}_{\text{covariance of } X, Y}$

covariance

$$\sigma_{XY} = \text{cov}(X,Y) = E[(X - E(X))(Y - E(Y))]$$
$$= E(XY) - E(X)E(Y)$$

Thrm: for independent r.v.'s X and Y,

$E(XY) = E(X)E(Y)$, so $\text{cov}(XY) = 0$

continuous case:
$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f(x,y)\,dx\,dy$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \underbrace{g(x) h(y)}_{\text{marginals}}\,dx\,dy$$
$$= \underbrace{\int_{-\infty}^{\infty} x g(x)\,dx}_{E(X)} \underbrace{\int_{-\infty}^{\infty} y h(y)\,dy}_{E(Y)}$$

Ex.

	$f(x,y)$	$x = -1$	0	1	$h(y)$
	-1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
y	0	$\frac{1}{8}$	0	$\frac{1}{8}$	$\frac{2}{8}$
	1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
	$g(x)$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{2}{8}$	1

$$E(X) = (-1)\left(\frac{3}{8}\right) + 0\left(\frac{2}{8}\right) + (1)\left(\frac{3}{8}\right)$$
$$= 0$$
$$E(XY) = (-1)(-1)\left(\frac{1}{8}\right) + (0)(-1)\left(\frac{1}{8}\right) - \frac{1}{8} + 0 + 0 + 0 - \frac{1}{8} + 0 + \frac{1}{8}$$
$$= 0$$

$\text{cov}(X,Y) = 0$

$f(x,y) \overset{?}{=} g(x)h(y)$

$f(0,0) = 0 \neq g(0)h(0) = \frac{2}{8}\cdot\frac{2}{8}$

X, Y not independent

properties of covariance

① $\text{cov}(X,X) = \text{var}(X) = E[XX] - E[X]E[X]$

② $\text{cov}(aX + b, cY + d) = ac\,\text{cov}(X,Y)$

③ $\text{cov}(X + Y, Z) = \text{cov}(X,Z) + \text{cov}(Y,Z)$

④ $\text{cov}(X,Y) = \text{cov}(Y,X)$

σ_{XY} is not scale-free, so its magnitude does not indicate the strength of the linear relation between X, Y.

Defn. correlation coefficient of X and Y is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Thrm: $\rho(aX + b, cY + d) = \rho(X,Y)$

Thus, correlation is scale-free.

proof:
$$\frac{\text{cov}(aX+b, cY+d)}{\sqrt{\text{var}(aX+b)\,\text{var}(cY+d)}} = \frac{ac\,\text{cov}(X,Y)}{\sqrt{a^2\text{var}(X)c^2\text{var}(Y)}}$$
$$= \frac{ac\,\text{cov}(X,Y)}{ac\sqrt{\text{var}(X)\text{var}(Y)}}$$

Thrm: $-1 \leq \rho(X,Y) \leq 1$

proof posted

Interpretation of ρ

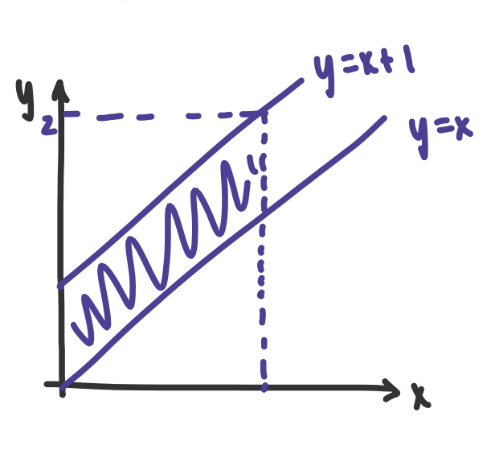
1) ρ is a measure of strength & direction of linear relation between X, Y

2) when there is an exact linear dependency

$\rho = 1$ or -1 $\quad Y = aX + b$

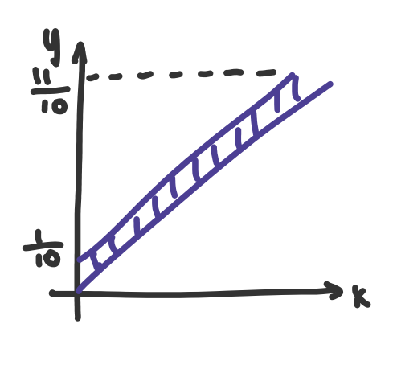
Ex. $f(x,y) = 1$ $\quad 0 < x < 1$

$x < y < x + 1$



$$E(X,Y) = \int_0^1 \int_x^{x+1} xy f(x,y)\,dy\,dx$$
$$= \frac{7}{12}$$
$$E(X) = \int_0^1 \int_x^{x+1} x f(x,y)\,dy\,dx$$
$$E(Y) = \int_0^1 \int_x^{x+1} y\,dy\,dx = 6$$

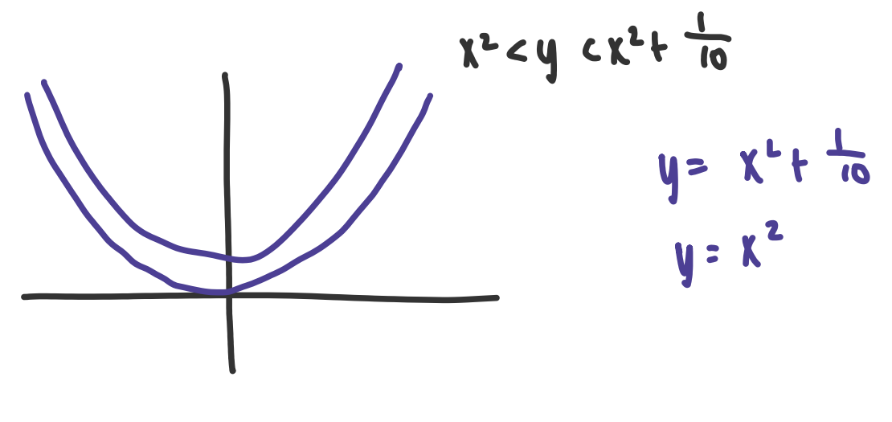
Ex. $f(x,y) = 10$, $0 < x < 1$, $x < y < x + \frac{1}{10}$



$$\rho = \sqrt{\frac{100}{101}}$$

Ex. $f(x,y) = 5$ $\quad -1 < x < 1$

$x^2 < y < x^2 + \frac{1}{10}$



strong relationship between X & Y

$\text{cov} = 0$

$$= E(X,Y) - E(X)E(Y)$$
$$= \int_{-1}^{1} \int_{x^2}^{x^2 + 1/10} xy\,5\,dy\,dx$$

Chapter 5: Discrete Distributions

Friday, February 6, 2026 11:14 AM

Discrete uniform: If the r.v. X assumes the values $x_1, \dots x_k$ with equal probabilities, then it has a discrete uniform dist.

$$f(x,k) = \frac{1}{k}, \quad x = x_1, x_2, \dots x_k$$
$$E(X) = \mu = \frac{1}{k}\sum_{i=1}^{k} x_i$$
$$\text{var}(X) = E[(X-\mu)^2] = \frac{1}{k}\sum_{i=1}^{k}(x_i-\mu)^2$$
$$= \sigma^2$$

Bernoulli Distribution

r.v. X has only 2 outcomes (success or failure)

$$X = \begin{cases} 1, & \text{success} \\ 0, & \text{failure} \end{cases}$$

x	0	1
$f(x)$	q	p

$$\sigma^2 = E(X^2) - \mu^2 = 0^2q + 1^2p - p^2$$
$$= p - p^2 = p(1-p) = pq$$

Ex. 4 components are tested independently. X: # of defectives

x	0	1	2	3	4
$P(X=x) = f(x)$	$\binom{4}{0}p^0q^4$	$\binom{4}{1}p^1q^3$	$\binom{4}{2}p^2q^2$	$\binom{4}{3}p^3q^1$	$\binom{4}{4}p^4q^0$

Binomial distribution: number of successes in n independent Bernoulli trials

$$b(x; n, p) = f(x) = \binom{n}{x} p^x q^{(n-x)}, \quad x = 0, 1, 2, \dots, n$$

$$\binom{n}{0}p^0q^n + \binom{n}{1}p^1q^{n-1} + \binom{n}{2}p^2q^{n-2} + \dots + \binom{n}{n}p^nq^0$$
$$= \underbrace{(p+q)}_{1}{}^{n}$$

← Binomial expansion

$$\mu = nE[I_i] = np$$
$$\text{var}(X) = \sigma^2 =$$
$$\text{var}(aX+b, cY+d) = a^2\text{var}(X) + c^2\text{var}(Y) + 2ac\,\text{cov}(X,Y)$$
$$\text{var}(X) = \text{var}(I_1 + I_2 + \dots + I_n)$$

$$E[X_1 + X_2 + \dots + X_n] = E(X_1) + E(X_2) + \dots + E(X_n)$$
$$X = \underbrace{I_1 + \dots + I_n}_{n \text{ independent Bernoulli variables}}$$

Ex. A manufacturer claims that their defective rate is 0.1. We test 4 components independently and find 3 defectives. What does this imply about the claim?

Assume $p = 0.1$

X = # of defectives in $n=4$ trials

$$P(X=x) = \binom{4}{x} 0.1^x\, 0.9^{4-x}, \quad x = 0, 1, 2, 3, 4$$
$$P(X \geq 3) = \binom{4}{3}0.1^3\,0.9 + \binom{4}{4}0.1^4$$
$$= 0.0037$$

← NOT "how likely is claim true"

Reject claim.

Multinomial distribution

If each trial has more than 2 possible outcomes

Defn: if a given trial can result in k outcomes $E_1, \dots E_k$, with probabilities $p_1, p_2, \dots p_k$, then the prob. distribution of r.v.'s $X_1, X_2, \dots X_k$, representing the number of occurrences for $E_1, E_2, \dots E_k$ in n independent trials is:

$$f(x_1, x_2, \dots x_k; p_1, p_2, \dots p_k, n) = \binom{n}{x_1, x_2, \dots x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

with $\sum_{i=1}^{k} x_i = n$ and $\sum_{i=1}^{k} p_k = 1$

Ex. As a student drives to school, he encounters a traffic light, which stays green for 35s, yellow for 5s, and red for 60s.

X_1 = # of times he encounters green in n trials
X_2 = " yellow "
X_3 = " red "

$X_1 + X_2 + X_3 = n$

$$\left.\begin{aligned} p(\text{green}) &= p_1 = 0.35 \\ p(\text{yellow}) &= p_2 = 0.05 \\ p(\text{red}) &= p_3 = 0.6 \end{aligned}\right\} p_1 + p_2 + p_3 = 1$$

$n = 100$:

$$P(X_1 = 30, X_2 = 10, X_3 = 60) = \binom{100}{30, 10, 60} 0.35^{30} \cdot 0.05^{10} \cdot 0.6^{60}$$
$$= \binom{100}{30}\binom{70}{10}\binom{60}{60}(0.00117)$$

$$\binom{n}{x_1, x_2, \dots x_n} = \binom{n}{x_1}\binom{n-x_1}{x_2}\dots\binom{n-x_1-x_2-\dots-x_{k-1}}{x_k} = \frac{n!}{x_1!x_2!\dots x_k!}$$

(1) Hypergeometric: sampling w/o replacement
trials are dependent

Ex. Hat contains 12 tickets, 7 black, 5 white. 3 are drawn at random.

X: number of black tickets drawn, $X = 0, 1, 2, 3$

$$P(X=2) = \frac{\binom{7}{2}\binom{5}{1}}{\binom{12}{3}}$$
$$P(X=x) = \frac{\binom{7}{x}\binom{5}{3-x}}{\binom{12}{3}}$$

Defn: pmf of hypergeometric r.v. X

The number of successes in a random sample of size n selected from N items of which k are labelled success and N-k labelled failure is

$$h(x; N, n, k) = f(x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}} \qquad \max\{0; n-(N-k)\} \leq x \leq \min\{n, k\}$$

Acceptance sampling

Ex. Company has inspection system for batches of small compressors, 15 each, unacceptable if it has 2 or more faulty. In inspection, a random sample of 5 is selected and all tested. If none is faulty, then batch is accepted. Is this a good plan?

Let's assume we have an unacceptable batch with 2 faulty.

$N = 15, \quad n = 5, \quad k = 2, \quad N-k = 13$

X: number of faulty in sample of 5

$$P(X = 0 \mid 2 \text{ faulty in batch}) = \frac{\binom{2}{0}\binom{13}{5}}{\binom{15}{5}} \approxeq 0.43$$

∴ not a good plan, should increase sample size

$$P(X=0 \mid 3 \text{ faulty}) = 0.26$$

$\mu = \frac{2}{3}$

Thrm: Mean & variance for hypergeometric

$$\mu = \frac{nk}{N} \qquad \sigma^2 = \frac{N-n}{N-1}\, n\, \frac{k}{N}\left(1 - \frac{k}{N}\right)$$

Relation to Binomial Dist.

If $n \ll N$ (rule of thumb: $\frac{n}{N} \leq 0.05$), the nature of N items changes very little w/ each draw. So a binomial dist. with $p = \frac{k}{N}$ can be used to approximate hypergeometric.

Ex. A tire manufacturer reports that among a shipment of 5000, 1000 are slightly blemished. If one purchases 10 of them at random, find prob 3 are blemished

$N = 5000 \qquad k = 1000$
$n = 10$

number of blemished in $n = 10$

$$P(X=3) = \frac{\binom{1000}{3}\binom{4000}{7}}{\binom{5000}{10}} = 0.2015$$

Binomial approx:

$$p = \frac{1000}{5000} = \frac{1}{5} = 0.2$$

$$P(X=3) \approxeq \binom{10}{3} 0.2^3\, 0.8^7 = 0.2013$$

Negative binomial dist:

Prob. that k^{th} success occurs on y^{th} trial

constant

Ex. Electronic component $p(D) = p = 0.1$

prob of 3^{rd} defective on 4^{th} trial

$$P(X < 4) = \binom{3}{2} 0.1^3\, 0.9$$
$$P(X = 5) = \binom{4}{2} 0.1^3\, 0.9^2$$

Defn: X: number of trials where the k^{th} success occurs in a sequence of independent Bernoulli trials

$$b^*(x; k, p) = f(x) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x = k, k+1, \dots$$
$$\mu = \frac{k}{p} \qquad \text{var} = \frac{k(1-p)}{p^2}$$

Geometric dist:

X: number of trial on which 1^{st} success occurs in a sequence of independent Bernoulli trials

negative binomial with $k = 1$

$$g(x; p) = pq^{x-1}, \quad x = 1, 2, 3, \dots$$
$$\mu = \frac{1}{p} \qquad \text{var} = \frac{1-p}{p^2}$$

Ex. sample components until we observe a D

$P(D) = 0.1$

x	$P(X=x)$
1	0.1
2	$0.9 \cdot 0.1$
3	$0.9^2 \cdot 0.1$
4	$0.9^3 \cdot 0.1$
⋮	⋮

Thrm: $X \sim$ geometric

for integers $s > t$

$$P(X > s \mid X > t) = P(X > s-t)$$

This is known as the "memoryless" property

The prob of getting an additional s-t failures having observed t failures already is the same as prob. of observing s-t failures from the start of the sequence.

Proof:

$$P(X > n) = P(\text{no success in } n \text{ trials}) = (1-p)^n = q^n$$
$$P(X > s \mid X > t) = \frac{P(X > s \text{ and } X > t)}{P(X > t)} = \frac{P(X > s)}{P(X > t)} = \frac{(1-p)^s}{(1-p)^t}$$
$$= (1-p)^{s-t} = P(X > s-t)$$

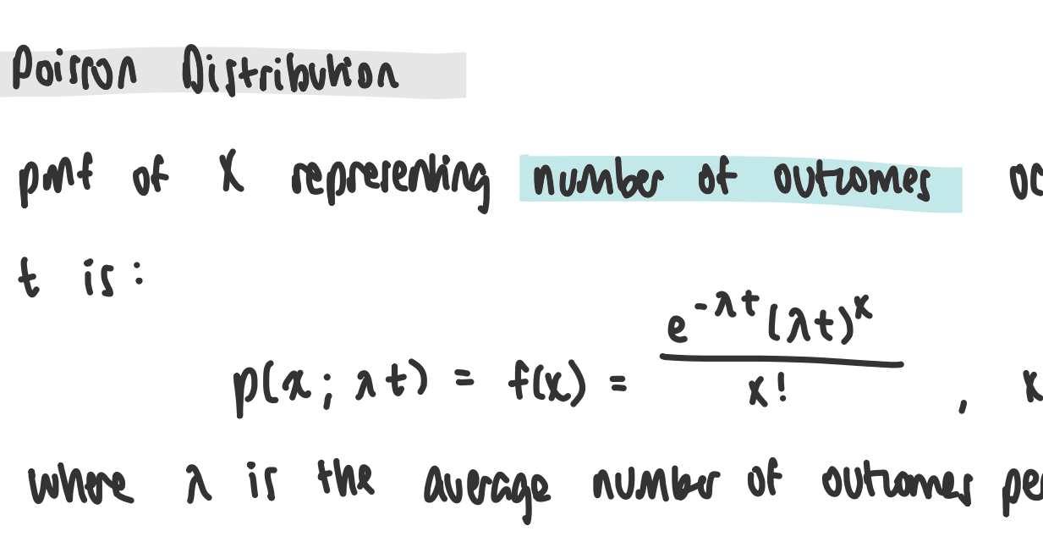
Poisson Distribution

pmf of X representing number of outcomes occurring in a given time interval or specified region denoted by t is:

$$p(x; \lambda t) = f(x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

where λ is the average number of outcomes per unit time, distance, area, volume, …

$$\mu = \sigma^2 = \lambda t$$



Ex. 7 is the average number of customers arriving each hour at a certain facility with $\lambda = 7$.

a) prob that more than 10 customers arrive in a 2h period

X: number of customers arriving in 2 hours

$$P(X > 10) = 1 - P(X \leq 10)$$
$$= 1 - \sum_{x=0}^{10} \frac{e^{-14} \cdot 14^x}{x!} = 0.8243$$

b) mean number of arrivals over a 2h period

$7 \times 2 = 14$

Poisson process

The basic form is a continuous counting process $\{N(t), t \geq 0\}$ that has the following properties

① $N(0) = 0$ → zero arrivals at the start

② **Independent increments:** # of outcomes occuring in one-time interval (or region) is independent of the # that occurs in any disjoint interval (or region)

③ **Stationary increments:** prob dist of # of occurrences in any time interval (or region) only depends on the length of interval

④ **no simultaneous outcomes:** prob that two outcomes occur in a very short interval is negligable

Thrm: $X \sim b(x; n, p)$

when $n \to \infty$, $p \to 0$, and $np \xrightarrow[n\to\infty]{} \mu$ remains constant,

$$b(x; n, p) \xrightarrow[n\to\infty]{} P(x; \mu)$$

We can use Poisson to approximate binomial for "rare occurrences"

Rule of thumb: Approximation is good for $n \geq 20$ and $p \leq 0.05$ or if $n \geq 100$, $np \leq 10$

Ex. On average, 1 in every 1000 glass products are blemished. Prob that a random sample of 8000 will yield fewer than 7 items that are blemished?

X: number of defectives in a sample of 8000

$\sim \text{binom}(n = 8000, \underset{\frac{1}{1000}}{p = 0.001})$

$$P(X < 7) = \sum_{x=0}^{6}\binom{8000}{x} 0.001^x\, 0.999^{8000} = 0.3132$$

sum(dbinom(0:6, 8000, 0.001))

$$\approx \sum_{x=0}^{6} P(x; 8) = 0.3134$$

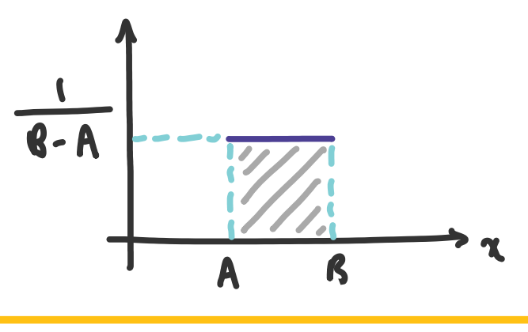
$$\frac{e^{-\lambda t}\overset{8}{(\lambda t)}{}^x}{x!}$$

Chapter 6: Continuous Probability Distribution

Wednesday, February 11, 2026 1:47 PM

Defn: pdf of uniform r.v. X on the interval $[A, B]$ is

$$f(x; A, B) = \begin{cases} \frac{1}{B-A}, & A \le x \le B \\ 0, & \text{otherwise} \end{cases}$$



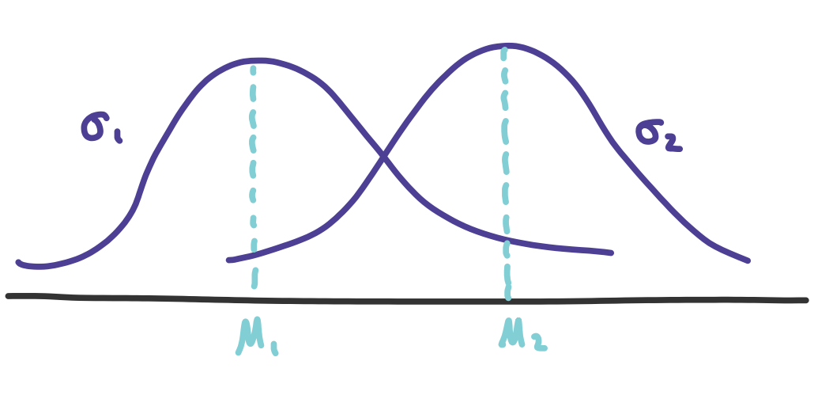
$$E(X) = \frac{A+B}{2}$$

$$\sigma^2 = \frac{(B-A)^2}{12}$$

Normal distribution

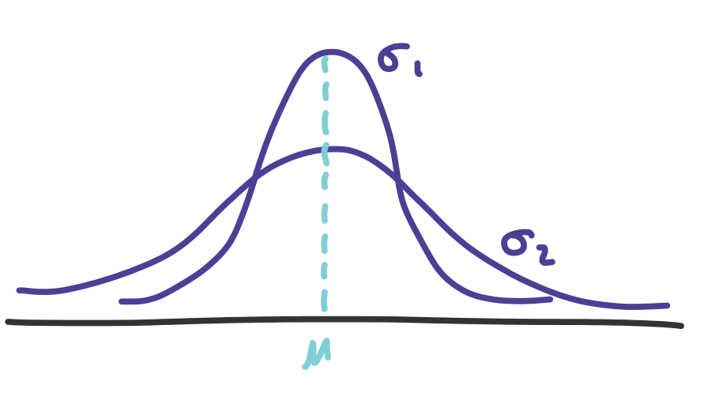
Defn: pdf of normal r.v. X with mean μ and variance σ^2 is

$$n(x; \mu, \sigma) = f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty$$



$\mu_1 < \mu_2$

$\sigma_1 = \sigma_2$



$\sigma_1 < \sigma_2$

Standardization, z-scores

Transform any normal r.v. X to a normal r.v. Z with mean 0 and variance 1

$$z = \frac{X-\mu}{\sigma} \qquad z_1 = \frac{x_1-\mu}{\sigma_1} \qquad z_2 = \frac{x_2-\mu}{\sigma_2}$$

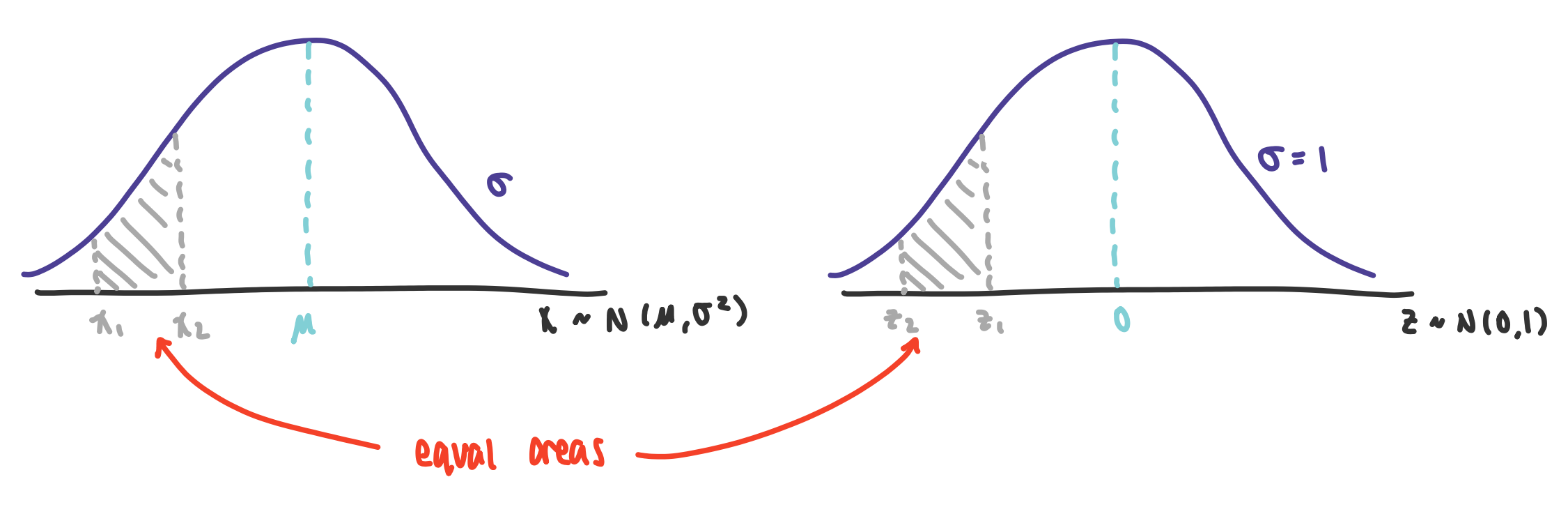
$$x = \sigma_z + \mu$$

$$dx = \sigma dz$$

$$P(x_1 \le X \le x_2) = \frac{1}{\sqrt{2\pi}\,\sigma} \int_{x_1}^{x_2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{1}{2}z^2} dx$$

$$= P(z_1 \le x \le z_2)$$



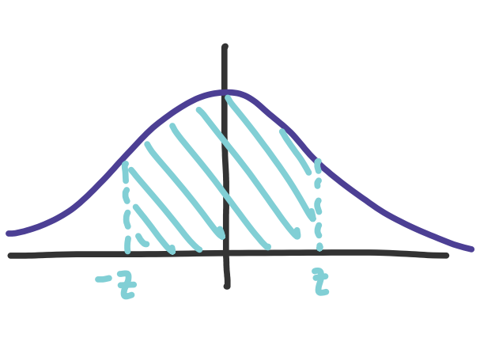
Defn: z-value or z-score tells us the how many std. deviations the original observation falls away from the mean $x = \mu + \sigma z$

Ex. $P(z < 1.4) = 0.9192$

$P(z > 1.4) = 1 - 0.9192 = 0.0808$

$P(z < -1.4) = 0.0808$

$P(0.43 < z < 2.15) = \underset{0.9798}{P(z < 2.15)} - \underset{0.6634}{P(z < 0.43)}$

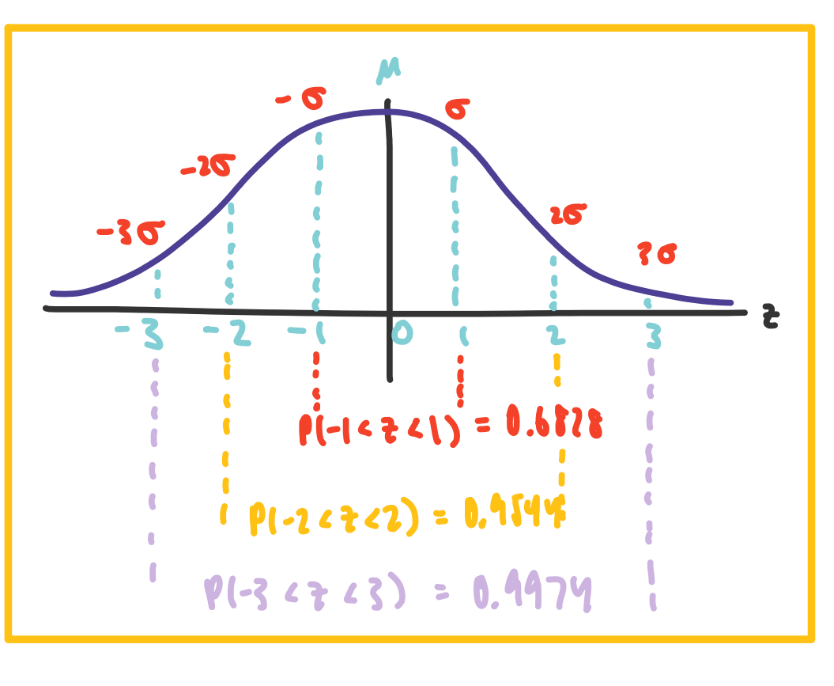


Ex. $X \sim N(\mu = 40, \sigma = 6)$

$P(X < \underset{?}{x}) = 0.45$

$P(z < \underbrace{\frac{x-40}{6}}_{\sim -0.13}) = 0.45$

$x = 40 - 0.13 \times 6 = 39.22$



Ex. An oil filter and change in ? min or next one free!

Time to complete $\sim N$ (15 min, $\sigma = 2.5$ min)

What number should be put in the ad if it was decided that about 5 in 100 customers is acceptable?

$P(X < x) = 0.95$

$P(\underset{z = 1.65}{z} < z) = 0.95$

$1.65 = \frac{x-15}{2.5}$

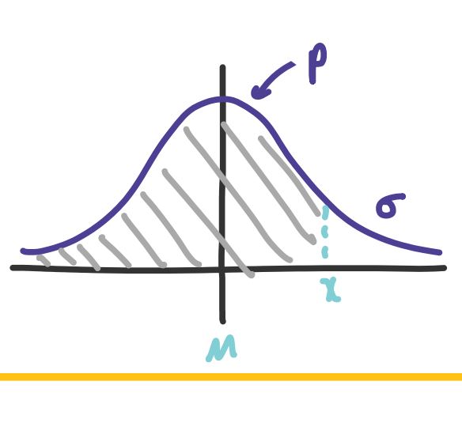
$x = 19.12$ min $\Rightarrow$ $x = 20$ min

Defn: $P(Z < z) = p$

z: p^{th} percentile of standard normal

$P(X < x) = p$

x: p^{th} percentile for r.v. X



Normal approximation to the binomial

If X is a binomial r.v. with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the dist. of

$$z = \frac{X - np}{\sqrt{npq}}$$

as $n \to \infty$, is the standard normal

$$P(X \le x) = \sum_{k=0}^{x} b(k; n, p) \approx P\left(z < \frac{x + 0.5 - np}{\sqrt{npq}}\right)$$

area under normal curve to the left of $x + 0.5$

Rule of thumb: use if $np \ge 5$ and $nq \ge 5$

Ex. 21% of children under age 6 live in households under poverty level. A study chooses random sample of 300 students.

a) mean of children from poverty? std dev?

X : number of poverty in 300 $\sim$ binom $(\underset{300}{n}\underset{0.21}{p}, \sqrt{np\underset{0.79}{q}} = \sigma)$

b) $P(X \ge 80) = \sum_{x=80}^{300} \binom{300}{x} 0.21^x 0.79^{n-x}$

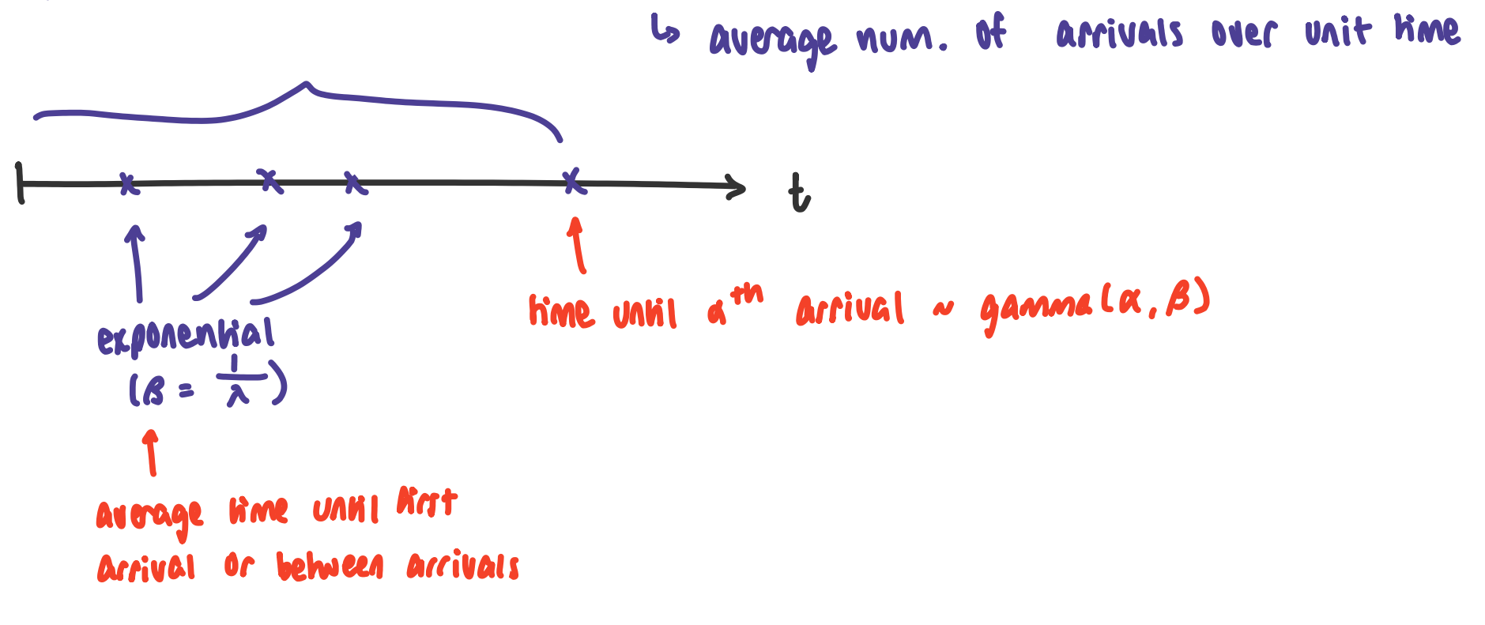
$= 1 - P(X \le 79)$

$= 1 - P\left(z \le \frac{79 + 0.5 - 63}{7.055}\right)$

$\approxeq 0.01$

# of arrivals in Poisson (λ)

↳ average num. of arrivals over unit time



t
exponential
$(\beta = \frac{1}{\lambda})$
average time until first arrival or between arrivals
time until α^{th} arrival $\sim$ gamma(α, β)

Gamma and Exponential Distribution

Defn: gamma function

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \quad \text{for } \alpha > 0$$

recursive : $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$

$\Gamma(n) = (n-1)!$ if n is integer

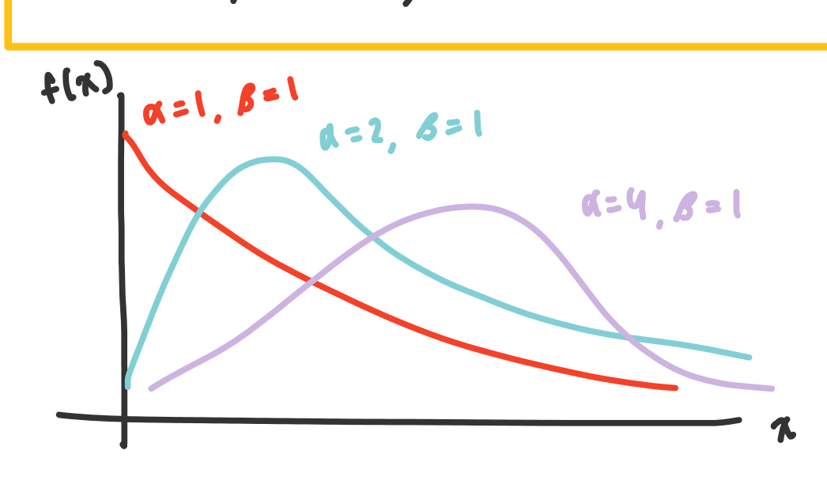
$\Gamma(\frac{1}{2}) = \sqrt{\pi}, \quad \Gamma(1) = 1$

Defn: cont. r.v. X has gamma dist. w/ parameters α and β

$$f(x; \alpha, \beta) = f(x) = \frac{1}{\beta^2 \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x > 0$$

where $\alpha, \beta > 0$

$\mu = \alpha\beta, \quad \sigma^2 = \alpha\beta^2$



If $\alpha = 1$, then exponential dist.

define $X \sim \exp(\beta)$

$$f(x; \beta) = f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0$$

where $\beta > 0$

$\mu = \beta, \quad \sigma^2 = \beta^2$

Relation to Poisson process

proof for time until first occurrence:

X = # of arrivals in t time

$\underset{\text{prob. of no arrivals at time t}}{P(X = 0, \lambda t)} = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}$

* poisson pdf: $\frac{e^{-\lambda t}(\lambda t)^x}{x!}, \quad x = 0, 1, 2, \ldots$

Let y be time until first poisson event.

$P(Y > y) = e^{-\lambda y}, \quad y > 0$

If $P(Y \le y) = 1 - e^{-\lambda y}, \quad y > 0$

pdf: $f(y) = \lambda e^{-\lambda y}, \quad y > 0$

take derivative wrt. y

exponential dist w/ $\lambda = \frac{1}{\beta}$

Ex. number of cars arriving at a certain intersection per minute has Poisson dist. with mean = 5

prob that more than 2 min are required before 10 cars arrive?

X: time until 10 events $\sim$ gamma $(\alpha = 10, \beta = \frac{1}{5})$

$P(X > 2) = 1 - P(X \le 2)$

$= 1 - \int_0^2 \frac{1}{\beta^2} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} dx$

change of variable : $y = \frac{x}{\beta}$ (so we can use Table A.23 in book)

$P(X \le 2) = P(Y \le 10) = \int_0^{10} \frac{y^{\alpha-1} e^{-y}}{\Gamma(\alpha)} dy$

$= \int_0^{10} \frac{y^9 e^{-y}}{\Gamma(10)} dy$

$= 0.542$

$y = \frac{x}{\beta}$

$\beta y = x$

$\beta dy = dx$

$P(X > 2) = 1 - 0.542 = 0.458$

Equipment failures often conform to Poisson process.

Ex. Life, in years, of a certain type of switch has an exp. dist w/ average life $\beta = 2$. If 100 many are installed in diff. systems, what's the prob that at most 30 will fail in the first year?

X: time until failure $\sim \exp(\beta = 2)$ exp. pdf: $\frac{1}{\beta} e^{-x/\beta}, \quad x > 0$

$P(X < 1) = \int_0^1 \frac{1}{2} e^{-x/2} dx$

$= 0.3935$

Y: number of switches that fail out of 100 within first year $\sim$ binomial $(n = 100, p = 0.3935)$

$P(Y \le 30) = \sum_{y=0}^{30} \binom{100}{y} 0.3935^y (1 - 0.3935)^{100-y}$ approximate as normal dist. (large n)

$\approx P\left(z \le \frac{30 + 0.5 - np}{\sqrt{npq}}\right)$

$= P(z < -1.81)$

$= 0.0352$

Thrm: Exponential dist. is memoryless

$$P(X \ge t) = P(X \ge t_0 + t \mid X \ge t_0)$$

If a component "makes it" to t_0 hours, the prob of lasting another t hours is the same as the prob of lasting t hours from the beginning

Weibull distribution

Defn: cont r.v. X has a weibull dist. w/ pdf

$$f(x; \alpha, \beta) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$F(x) = 1 - e^{-\alpha x^\beta}, \quad x \ge 0$

$\mu = \alpha^{-1/\beta} \Gamma(1 + \frac{1}{\beta}) \qquad \sigma^2 = \alpha^{-2/\beta} \{\Gamma(1 + \frac{2}{\beta}) - [\Gamma(1 + \frac{1}{\beta})^2]\}$

Failure rate (or hazard rate)

Define reliability as "the prob that a product will function properly for at least a specified time under specified conditions"

$R(t)$: reliability at time t $= P(\underset{\text{time until failure}}{\underline{T}} > t) = \int_t^\infty \underset{\text{pdf of T}}{f(t)} dt = 1 - \underset{\text{cdf of T}}{F(t)}$

conditional prob that a component will fail in the interval $T = t$ to $T = t + \Delta t$, given it survived until t

$$\frac{P(t < T < t + \Delta t)}{P(t < T)} = \frac{F(t + \Delta t) - F(t)}{R(t)}$$

Failure rate: change over time of the above conditional probability

$$z(t) = \lim_{\Delta t \to 0} \frac{\frac{F(t + \Delta t) - F(t)}{R(t)}}{\Delta t} = \frac{F'(t)}{R(t)} = \frac{f(t)}{R(t)} = \frac{f(t)}{1 - F(t)}$$

failure rate for weibull at time t

$z(t) = \alpha\beta t^{\beta-1}, \quad t > 0 \qquad \alpha, \beta > 0$

① If $\beta = 1$, then $z(t) = \alpha$, we get exponential

② If $\beta > 1$, then $z(t)$ is increasing function of t → component wears over time

③ If $\beta < 1$, then $z(t)$ is a decreasing function of t → component strengthens over time

Chi-squared dist

special case of gamma dist. with $\alpha = \frac{\gamma}{2}$ and $\beta = 2$

$\mu = \gamma \qquad \sigma^2 = 2\gamma$

γ is a pos. integer called degrees of freedom

Chapter 7: Transformations, Moment-generating functions

Friday, February 27, 2026 3:11 PM

finding pmf/pdf of functions of random variables

Thrm: suppose X is a discrete r.v. with pmf $f(x)$

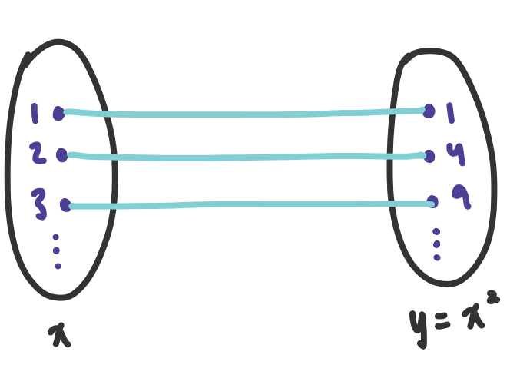
Let $Y = U(X)$ define a 1-to-1 transformation so that $y = u(x)$ can be uniquely solved for $x = w(y) = u^{-1}(y)$

Then, pmf of Y is $g(y) = f(w(y))$

Ex. $f(x) = \frac{3}{4}\left(\frac{1}{4}\right)^{x-1}$, $x = 1, 2, 3, \dots$ geometric

pmf of $Y = X^2$:

$y = x^2$ $x = \sqrt{y}$ $g(y) = f(\sqrt{y}) = \frac{3}{4}\left(\frac{1}{4}\right)^{\sqrt{y}-1}$, $y = 1, 4, 9, \dots$



$P(X=1) = P(Y=1)$

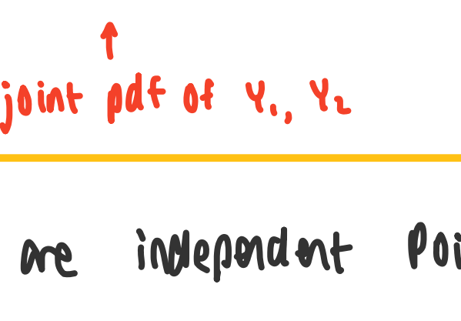
$P(X=2) = P(Y=4)$

$P(X=3) = P(Y=9)$

Ex. If not 1-to-1

$Y = X^2$

$$g(y) = \begin{cases} P(X=\sqrt{y}) + P(X=-\sqrt{y}), & y > 0 \\ P(X=0), & y = 0 \\ 0, & y < 0 \end{cases}$$



Thrm: suppose X_1 and X_2 are discrete r.v.'s with joint pmf $f(x_1, x_2)$

Let $Y_1 = U_1(X_1, X_2)$ and $Y_2 = U_2(X_1, X_2)$ be 1-to-1 transformation with $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ can be uniquely solved as $x_1 = w_1(y_1, y_2)$ and $x_2 = w_2(y_1, y_2)$, then:

$$g(y_1, y_2) = f[w_1(x_1, x_2), w_2(x_1, x_2)]$$

joint pdf of Y_1, Y_2

Ex. X_1, X_2 are independent Poisson distributed r.v.'s with means μ_1, μ_2, respectively. Find pmf of $Y = X_1 + X_2$

$$f(x_1, x_2) = \frac{e^{-\mu_1}\mu_1^{x_1}}{x_1!}\,\frac{e^{-\mu_2}\mu_2^{x_2}}{x_2!} \qquad x_1, x_2, = 0, 1, 2, \dots$$

$Y_1 = X_1 + X_2$, $Y_2 = X_2$

$x_1 = y_1 - y_2$, $x_2 = y_2$

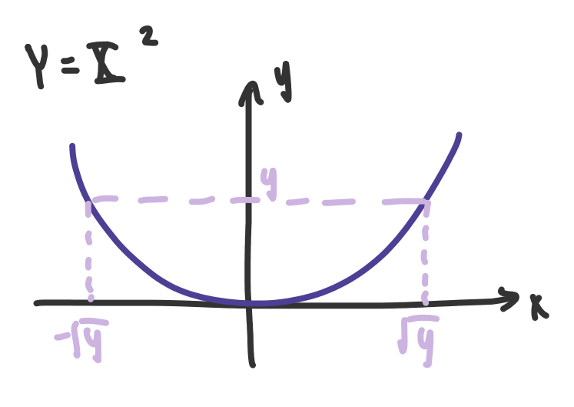
$$g(y_1, y_2) = \frac{e^{-\mu_1}\mu_1^{y_1-y_2}}{(y_1-y_2)!}\,\frac{e^{-\mu_2}\mu_2^{y_2}}{y_2!}$$

$$h(y) = \frac{e^{-(\mu_1+\mu_2)}}{y_1!} \underbrace{\sum_{y_2=0}^{y_1} \frac{y_1!}{y_2!(y_1-y_2)!}\mu_1^{y_1-y_2}\mu_2^{y_2}}_{\text{binomial expansion of } (\mu_1+\mu_2)^{y_1}}$$

$$= \frac{e^{-(\mu_1+\mu_2)}(\mu_1+\mu_2)^{y_1}}{y_1!}$$

Ex. suppose $X \sim \text{uni}(0,1)$

let $Y = X^2$, find cdf of Y



$P(Y \le y) = P(X^2 \le y) = P(X \le \sqrt{y})$

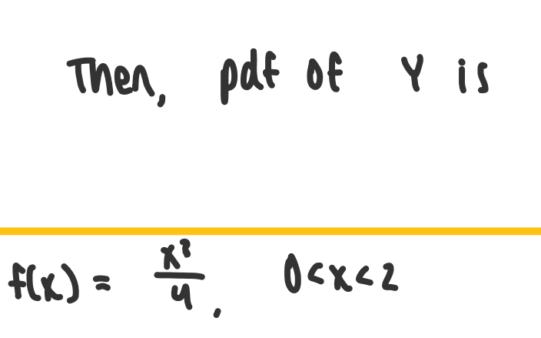
$P(X \le x) = \int_0^x 1\,dx = x$

$P(Y \le y) = \sqrt{y}$, $0 < y < 1$

derivative $f(y) = \frac{1}{2\sqrt{y}}$

Ex. $f(x) = \frac{x+1}{2}$, $-1 \le x \le 1$

$Y = X^2$



$P(Y \le y) = P(-\sqrt{y} \le X \le \sqrt{y})$

$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{x+1}{2}\,dx$

$= \sqrt{y}$, $0 \le y \le 1$

Thrm: X is cont r.v. w/ pdf $f(x)$.

Let $Y = U(X)$ define a 1-to-1 transformation that can be uniquely solved as $x = w(y)$

Then, pdf of Y is $g(y) = f[w(y)]\,|J|$

$J = w'(y) \to$ Jacobian of transformation

Ex. $f(x) = \frac{x^3}{4}$, $0 < x < 2$

$Y = X^6$

$P(Y \le y) = P(X^6 \le y) = P(X \le \sqrt[6]{y})$

$P(X \le x) = \int_0^x \frac{x^3}{4}\,dx = \frac{x^4}{16}$

$P(Y \le y) = \frac{(\sqrt[6]{y})^4}{16}$

$= \frac{y^{2/3}}{16}$

$g(y) = F'(y) = \frac{y^{-\frac{1}{3}}}{24}$, $0 < y < 64$

Jacobian:

$x = y^{1/6} = w(y) \Rightarrow w'(y) = \frac{1}{6}y^{-\frac{5}{6}} = J$

$g(y) = \frac{(y^{1/6})^3}{4}\left|\frac{1}{6}y^{-\frac{5}{6}}\right|$, $0 < y < 64$

$= \frac{y^{-1/3}}{24}$, $0 < y < 64$

Thrm: X_1, X_2 are cont. r.v.'s with joint pdf $f(x_1, x_2)$

Let $Y_1 = U_1(X_1, X_2)$, $Y_2 = U_2(X_1, X_2)$ be 1-to-1 transformation so that solutions $x_1 = w_1(y_1, y_2)$, $x_2 = w_2(y_1, y_2)$ are unique.

Then joint pdf:

$$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)]\,|J|$$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Ex. X_1, X_2 ind. exp. r.v.'s w/ mean β.

$Y = X_1 + X_2$. find pdf?

say $Y_2 = X_1$

$f(x_1, x_2) = \frac{1}{\beta}e^{-x_1/\beta} \cdot \frac{1}{\beta}e^{-x_2/\beta}$, $x_1, x_2 > 0$

$= \frac{1}{\beta^2}e^{-(x_1+x_2)/\beta}$, $x_1, x_2 > 0$

$x_1 = y_2$

$x_2 = y_1 - y_2$

$J = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$

$g(y_1, y_2) = \frac{1}{\beta^2}e^{-y_1/\beta}$, $0 < y_2 < y_1$

integrate $g(y_1) = \int_0^{y_1} \frac{1}{\beta^2}e^{-y_1/\beta}\,dy_2 = \frac{1}{\beta^2}e^{-y_1/\beta}y_1$, $0 < y_1$

↳ gamma r.v. $\alpha = 2, \beta$

Thrm: suppose X is a cont r.v. w/ pdf $f(x)$.

Let $u = Y(X)$ be NOT a 1-to-1 transformation.

If the interval over which x is defined can be partitioned into k mutually disjoint sets such that each inverse function $x_1 = w_1(y), \dots x_k = w_k(y)$ of $y = u(x)$ defines a 1-to-1 correspondence, then pdf of Y is

$$g(y) = \sum_{i=1}^{k} f[w_i(y)]\,J_i \quad \text{where } J_i = w_i(y),\ i = 1, \dots k$$

Ex. show that $Y = \frac{(X-\mu)^2}{\sigma^2} \sim \chi^2(1)$ when $X \sim N(\mu, \sigma^2)$

In other words, $Y = Z^2$, $Z \sim N(0,1)$

$f(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$, $-\infty < z < \infty$

$y = z^2$, $z = \pm\sqrt{y}$ $\begin{cases} z_1 = -\sqrt{y}, & z < 0 \\ z_2 = \sqrt{y}, & z > 0 \end{cases}$ $J_1 = -\frac{1}{2\sqrt{y}}$ $J_2 = \frac{1}{2\sqrt{y}}$

$g(y) = \frac{1}{\sqrt{2\pi}}e^{-y/2}\left|-\frac{1}{2\sqrt{y}}\right| + \frac{1}{\sqrt{2\pi}}e^{-y/2}\left|\frac{1}{2\sqrt{y}}\right|$

$= \frac{1}{\sqrt{2\pi}}y^{1/2-1}e^{-y/2}$, $y > 0$, gamma$(\alpha = \frac{1}{2}, \beta = 2) \sim \chi^2(1)$

Moment & moment generating functions (mgf's)

Defn: the r^{th} moment about zero of the r.v. X is

$$\mu_r' = E(x^r) = \begin{cases} \sum x f(x), & x \text{ disc.} \\ \int_{-\infty}^{\infty} x f(x)\,dx, & x \text{ cont.} \end{cases}$$

first moment: $E(X) = \mu$

second moment: $E(X^2) = \sigma^2 + \mu^2$

Defn: central moment (moment around μ)

$E[(X-\mu)^r]$ 3^{rd} moment: skewness

$E[(X-\mu)^2] = \sigma^2$ 4^{th} moment: kurtosis

Defn: mgf of X, $M_X(t) = E(e^{tX}) = \begin{cases} \sum_x e^{tx} f(x), & x \text{ disc.} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx, & x \text{ cont.} \end{cases}$

mgf exist if $\sum$ or $\int$ converges

Thrm: $\left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} = \mu_r'$

Ex. $X \sim \exp(\beta)$ $f(x) = \frac{1}{\beta}e^{-x/\beta}$, $x > 0$

$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx}\frac{1}{\beta}e^{-x/\beta}\,dx$

$= \frac{1}{\beta}\left.\frac{e^{(t-\frac{1}{\beta})x}}{t-\frac{1}{\beta}}\right|_0^\infty$

If $t < \frac{1}{\beta}$ $M_X(t) = \frac{1}{1-\beta t}$

mean: $\left.\frac{dM_X(t)}{dt}\right|_{t=0} = \mu_1' = \left.\frac{d(1-\beta t)^{-1}}{dt}\right|_{t=0} = \beta$

var: $\left.\frac{d^2 M_X(t)}{dt^2}\right|_{t=0} = 2\beta^2 = E(X^2)$

$\sigma^2 = E(X^2) - \mu^2 = 2\beta^2 - \beta^2 = \beta^2$

Thrm: Let X and Y be r.v.'s with mgf's $M_X(t)$ and $M_Y(t)$

If $M_X(t) = M_Y(t)$ for all t, then X and Y have same dist.

properties of mgf

① $Y = a + bX$, $a, b \in \mathbb{R}$, then $M_Y(t) = e^{at}M_X(bt)$

proof: $M_Y(t) = E(e^{tY}) = E(e^{ta+tbX})$

② If $X_1, \dots X_n$ ind. r.v.'s w/ mgf's $M_{X_1}(t), \dots M_{X_n}(t)$, and $Y = X_1 + \dots + X_n$, then $M_Y(t) = M_{X_1}(t)M_{X_2}(t)\dots M_{X_n}(t)$

proof: cont. case

$y(t) = E(e^{tY}) = E(e^{t(X_1+\dots+X_n)}) = \int_{x_n}\cdots\int_{x_1} e^{t(x_1+\dots+x_n)} f_1(x_1)f_2(x_2)\dots f_n(x_n)\,dx_1\dots dx_n$

$= \int_{x_1} e^{tx_1}f_1(x_1)\,dx_1 \cdot \int_{x_2} e^{tx_2}f_2(x_2)\,dx_2 \dots \int_{x_n} e^{tx_n}f_n(x_n)\,dx_n = M_{X_1}(t)\dots M_{X_n}(t)$

*cheat sheet

Ex. $X_1 \sim \text{Pois}(\lambda_1)$, $X_2 \sim \text{Pois}(\lambda_2)$ } ind. $Y = X_1 + X_2 \sim$? $\text{Pois}(\lambda_1 + \lambda_2)$

$M_{X_1}(t) = E(e^{tX_1}) = e^{\lambda_1(e^t-1)}$

$M_Y(t) = M_{X_1}(t)M_{X_2}(t) = e^{(\lambda_1+\lambda_2)(e^t-1)}$

Poisson $(\lambda_1 + \lambda_2)$

Thrm: $X \sim N(\mu, \sigma^2)$ $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$

Thrm: $X \sim \chi^2(\nu)$ $M_X(t) = (1-2t)^{-\nu/2}$

Thrm: If $X_1, \dots X_n$ are ind. r.v.'s w/ normal dist w/ means $\mu_1, \dots \mu_n$ of variance $\sigma_1^2, \dots \sigma_n^2$, then $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n \sim N(a_1\mu_1 + \dots + a_n\mu_n, a_1^2\sigma_1^2 + \dots + a_n^2\sigma_n^2)$

proof: $Y = a_1X_1 + a_2X_2$

$M_X(t) = M_{X_1}(a_1t) \cdot M_{X_2}(a_2t)$

$= \exp(a_1\mu_1 t + a_1^2\sigma_1^2\frac{t^2}{2} + a_2\mu_2 t + a_2^2\sigma_2^2\frac{t^2}{2})$

$= \exp[(a_1\mu_1 + a_2\mu_2)t + (a_1^2\sigma_1^2 + a_2^2\sigma_2^2)\frac{t^2}{2}]$

mgf of normal

special cases

① If $X_1 \sim N(\mu, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$, $X_n \sim N(\mu_n, \sigma_n^2)$ and $X_1, \dots X_n$ ind.

$X_1 + X_2 + \dots + X_n \sim N(\mu_1 + \dots + \mu_n, \sigma_1^2 + \dots + \sigma_n^2)$

① If $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

independently identically dist

$Y = X_1 + \dots + X_n \sim N(n\mu, n\sigma^2)$

$\bar{Y} = \frac{1}{n}X_1 + \dots + \frac{1}{n}X_n \sim N(\mu, \frac{\sigma^2}{n})$

Thrm: $X_1, \dots X_n$ χ^2 r.v.'s with dist. $\nu_1, \dots \nu_n$

independent

$Y = X_1 + \dots + X_n \sim \chi^2(\nu_1 + \dots + \nu_n)$

Corollary: $X_1, \dots X_n$ iid $N(\mu, \sigma^2)$ then

$Y = \sum_{i=1}^{n}\left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(n)$ recall $\left(\frac{X-\mu}{\sigma}\right) \sim \chi^2(1)$

$Y = \sum_{i=1}^{n}\left(\frac{X_i - \mu_i}{\sigma_i}\right) \sim \chi^2(n)$

$X_i \sim$ independent $N(\mu_i, \sigma_i^2)$

Chapter 8: Fundamental sampling distributions and data descriptions

Tuesday, March 3, 2026 3:17 PM

population: totality of observations which we are interested

sample: subset of a population

parameter: number that describes the population. Fixed but in practice unknown

statistic: a function of sample data. It is a r.v. w/ a distribution

Defn: The r.v.'s $X_1, \dots, X_n$ are said to form a (simple) random sample of size n if the X_i's are iid (independent, identically distributed) w/ each X_i having prob. dist. $f(x)$

$x_1 \ \vdots \ x_n$, $\bar{x}$ $\quad$ $x_{11} \ \vdots \ x_{1n}$, $\bar{x}_1$ $\quad$ $x_{21} \ \vdots \ x_{2n}$, $\bar{x}_2$

$X_1 \sim f(x_1)$
$X_2 \sim f(x_2)$
$\vdots$
$X_n \sim f(x_n)$

Defn: prob dist. of a statistic is called a sampling dist.

Sampling dist depends on size of population, size of sample, and method of sampling

Central Limit Theorem:
$X_1, \dots X_n$ independent from same dist. w/ mean μ and var. σ^2. Then, the limiting form of dist of
$$z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \quad \text{as } n \to \infty \quad \text{is the standard normal}$$
good for $n \geq 30$

① Sample from normal, $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$

then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ (s rough mgf's)

suppose a r.v. of n is taken from $N(\mu, \sigma^2)$. Each observation X_i, $i = 1, \dots n$ have the same dist.

② CLT → X_i : iid from a dist w/ mean μ and var σ^2

$\bar{X} \overset{app}{\sim} N(\mu, \frac{\sigma^2}{n})$

$n \geq 30$ is a good cutoff for the approximation

Thrm: If $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, then
$$\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2)$$

proof: normal mgf

$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$ — $N(\mu, \sigma^2)$

$M_{\sum_{i=1}^n X_i}(t) = M_{\bar{X}}(nt)$

$= \exp(n\mu t + n^2 \frac{\sigma^2}{n} \frac{t^2}{2})$

$\downarrow$

$N(n\mu, n\sigma^2)$

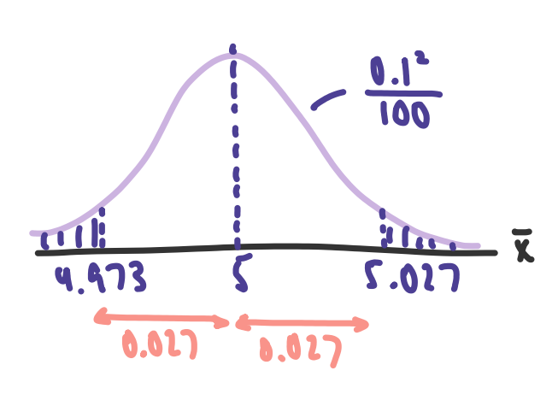
Inverses on the population mean

Ex. A manufacturing process provides cylindrical coordinates. The engineer involved conjectures that population mean diameter is 5mm. 100 parts are randomly selected. $\bar{x} = 5.027$mm ← sample average

we know that $\sigma = 0.1$. Does this sample support engineer's conjecture?

X_i : diameter of a part $\sim (\mu, 0.1^2)$

$\bar{X} \overset{app}{\sim} N(\mu, \frac{0.1^2}{100})$ $\quad$ assume $\mu = 5$mm



$\mu = 5 : H_0$
$\mu \neq 5 : H_a$

$P(\bar{X} < 4.973) + P(\bar{X} > 5.027)$

$2P(z > \frac{5.027-5}{\frac{0.1}{10}}) = 0.007$ ← refutes H_0

Sampling dist. of the difference between two averages

Thrm: If independent samples of size n_1 and n_2 are drawn at random from two populations, with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively and n_1 and n_2 are sufficiently large
$$\bar{X}_1 - \bar{X}_2 \overset{app}{\sim} N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$
proof: $\bar{X}_1 \overset{app}{\sim} N(\mu_1, \frac{\sigma_1^2}{n_1})$, $\bar{X}_2 \overset{app}{\sim} N(\mu_2, \frac{\sigma_2^2}{n_2})$

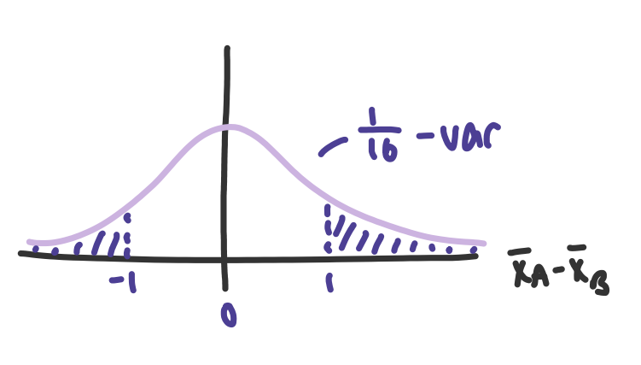
Ex. 32 specimens are painted w/ paint A, another 32 w/ paint B

Drying time in hours recorded:

$\sigma = 1$ $\quad$ $\bar{X}_A - \bar{X}_B = 1$

Is there a difference between μ_A and μ_B?

$\bar{X}_A - \bar{X}_B \sim N(\mu_A - \mu_B, \underbrace{\frac{1}{32} + \frac{1}{32}}_{\text{var}})$



$P(\bar{X}_A - \bar{X}_B < -1) + P(\bar{X}_A - \bar{X}_B > 1) = 2P(z > \frac{1-0}{\frac{1}{4}}) < 0.0001$

refute $\mu_A - \mu_B$

Law of Large Numbers

X_1, X_2 w/ mean μ and var σ^2

$\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$ $\quad$ as $n \to \infty$ $\quad$ $\underbrace{\text{var}(\bar{X}_n)}_{\frac{\sigma^2}{n}} \to 0$

$E(\bar{X}_n) = \mu$

so $\bar{X}_n \to E(\bar{X}_n) = \mu$

more formally,

weak law of large numbers

Let $X_1, \dots X_n$ be iid. r.v.'s with $E(X_i) = \mu$ and $\text{var}(X_i) = \sigma^2 < \infty$

Then for every $\varepsilon > 0$,
$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| < \varepsilon) = 1$$
ie. $\bar{X}_n$ converges to μ in probability

Strong Law of large numbers

$\underbrace{P(\lim_{n\to\infty} |\bar{X}_n - \mu| < \varepsilon) = 1}_{\text{happen for sure}}$ $\quad$ almost sure convergence

Thrm: Chebyshev's Inequality

prob. that X assumes a value within k std. devs of mean:

$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$ $\quad$ $k = 2$, $1 - \frac{1}{4} = \frac{3}{4}$

Thrm: Markov's inequality

If X is a non-negative r.v. w/ $E(X) < \infty$ and $a > 0$, then $P(X \geq a) \leq \frac{E(X)}{a}$

↳ no more than $\frac{1}{5}$ of pop. can have more than 5 times the average income

proof cont. case: $E(X) = \int_0^\infty x f(x)dx = \int_0^a x f(x)dx + \int_a^\infty x f(x)dx \geq \int_a^\infty x f(x)dx \geq \int_a^\infty a f(x)dx = aP(X \geq a)$

Sampling dist of S^2

Thrm: If S^2 is the variance of a random sample of size n, taken from a normal dist. having variance σ^2, then
$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n-1)$$
↳ ν degrees of freedom

proof: $\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n [(X_i - \bar{X}) + (\bar{X} - \mu)]^2$

$= \sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (\bar{X} - \mu)^2 + 2(\bar{X} - \mu)\underbrace{\sum_{i=1}^n (X_i - \bar{X})}_{0}$

$= \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$

divide by σ^2, substitute $(n-1)S^2$ w/ $\sum_{i=1}^n (X_i - \bar{X})^2$

$\underbrace{\frac{1}{\sigma^2}\sum_{i=1}^n (X_i - \bar{X})^2}$ $= \frac{(n-1)S^2}{\sigma^2} + \frac{(\bar{X} - \mu)^2}{\frac{\sigma^2}{n}}$

sum of n independent $\chi^2(1)$ r.v.'s

$X_i \overset{iid}{\sim} N(\mu, \sigma^2)$

$\frac{X_i - \mu}{\sigma} \sim N(0,1)$

$(\frac{X_i - \mu}{\sigma})^2 \sim \chi^2(1)$

↳ $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$

Thrm: If $X_1 \sim \chi^2(\nu_1)$, $X_2 \sim \chi^2(\nu_2)$, … $X_k \sim \chi^2(\nu_k)$ are independent, then
$$\sum_{i=1}^k X_i \sim \chi^2(\nu_1 + \dots + \nu_k)$$
can be easily proven w/ χ^2 mgf.

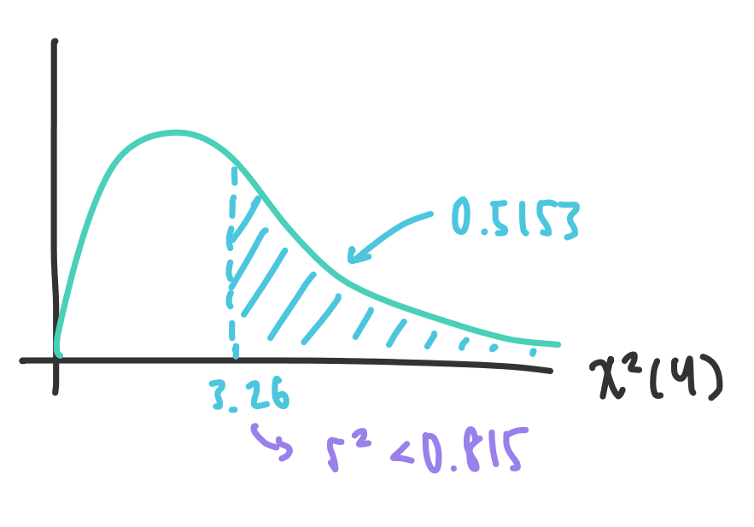
Ex. manufacturer guarantees its car batteries will last on average 3 years w/ σ at most 1 year. 5 are tested and found to have lifetimes 1.9, 2.4, 3.0, 3.5, and 4.2 years. Is σ claim correct?

$s^2 = 0.815$

Then, $\chi^2 = \frac{\overset{n-1}{(4)}\overset{s^2}{(0.815)}}{\underset{\sigma^2}{1}} = 3.26$

Assume $\sigma^2 = 1$

Assume lifetime normal



fail to reject

$\sigma^2 < 1$

Thrm: Let $z \sim N(0,1)$ and $V \sim \chi^2(\nu)$ and z and V independent.

Then, r.v. $T = \frac{z}{\sqrt{V/\nu}}$ has pdf
$$h(t) = \frac{\Gamma[(\nu+1)/2]}{\Gamma(\nu/2)\sqrt{\pi\nu}}\left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \quad -\infty < t < \infty$$
→ t dist. w/ ν degrees of freedom

Thrm: Let $X_1, \dots, X_n$ be ind. r.v.'s that are normal w/ mean μ, var σ^2

Then, $T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$

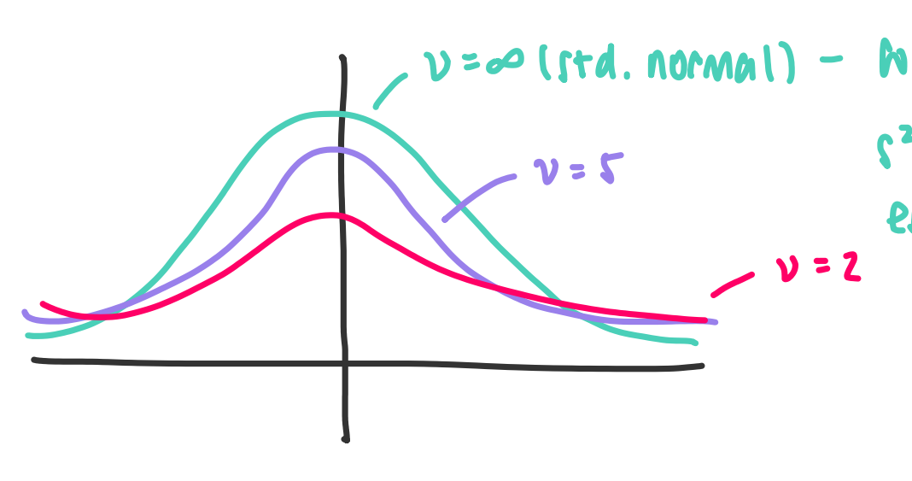
proof: $z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$\bar{X}$ & S^2 are ind., so are z and V.

Useful tips:

① If n is small, s fluctuates from sample groups to sample groups significantly so compared to $z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$, $T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ has more variability.

Thus, t-dist has wider tails.

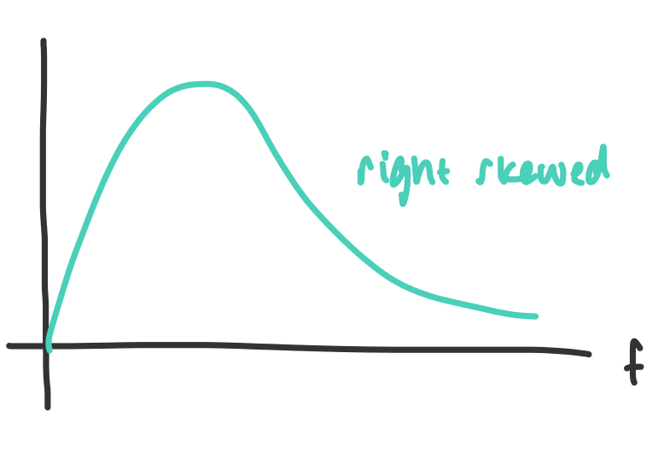


— for practical purposes, $n \geq 30$ s^2 becomes sufficiently good estimator of σ^2

F-dist

Thrm: Let $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$ be independent

$F = \frac{U/\nu_1}{V/\nu_2}$ has pdf
$$h(f) = \frac{\Gamma[(\nu_1+\nu_2)/2](\nu_1/\nu_2)^{\nu_1/2}}{\Gamma(\nu_1/2)\Gamma(\nu_2/2)} \frac{f^{(\nu_1/2)-1}}{(1+\nu_1 f/\nu_2)^{(\nu_1+\nu_2)/2}}, \quad f > 0$$



Thrm: If S_1^2 & S_2^2 are variances of ind. random samples of size n_1 and n_2, taken from normal populations w/ variances σ_1^2 and σ_2^2, respectively, then
$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

proof: $U = \frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1 - 1)$

$V = \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2 - 1)$

Thrm: $f_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{f_\alpha(\nu_2, \nu_1)}$

Review: population, sample, parameter, statistic

sampling dist., law of large numbers, CLT

① $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
② $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, then $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$
③ $X_{1i} \overset{iid}{\sim} N(\mu_1, \sigma_1^2)$ and $X_{2i} \overset{iid}{\sim} N(\mu_2, \sigma_2^2)$ independent, then
$\bar{X}_1 - \bar{X}_2 \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$

Chapter 9: One and Two Sample Estimation

Friday, March 13, 2026 3:04 PM

Defn: a point estimate of some population param. θ is a single value, $\hat{\theta}$ of statistic $\hat{\Theta}$
$\uparrow$ e.g. μ $\uparrow$ e.g. $\bar{x}$ $\uparrow$ e.g. $\bar{X}$

There will be some uncertainty in the estimate.

Ex. To estimate μ we can use $\bar{X}$ or $\tilde{X}$ (median). How do we know what is a better estimate?

Defn: A statistic $\hat{\Theta}$ is said to be an unbiased estimator of the parameter θ if

$$\mu_{\hat{\Theta}} = E(\hat{\Theta}) = \theta$$

we want the sampling dist. of $\hat{\Theta}$ to have an expected value equal to θ (ie. the parameter estimated)

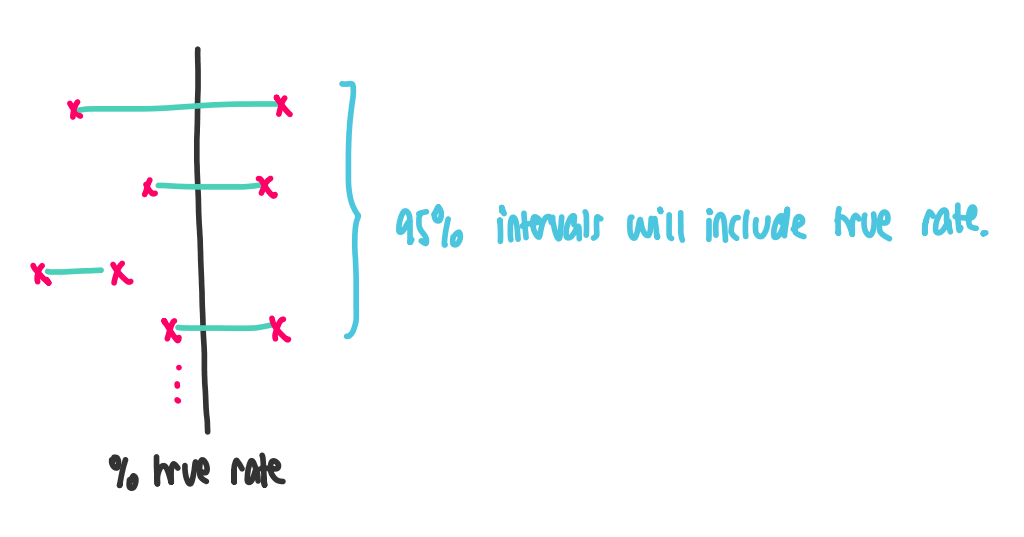
Thrm: $E(\bar{X}) = \mu$

proof: $E\left(\frac{X_1}{n} + \frac{X_2}{n} + \dots + \frac{X_n}{n}\right) = \frac{1}{n}E(X_1) + \frac{1}{n}E(X_2) + \dots + \frac{1}{n}E(X_n) = n\frac{\mu}{n} = \mu$

Thrm: $E(s^2) = \sigma^2$

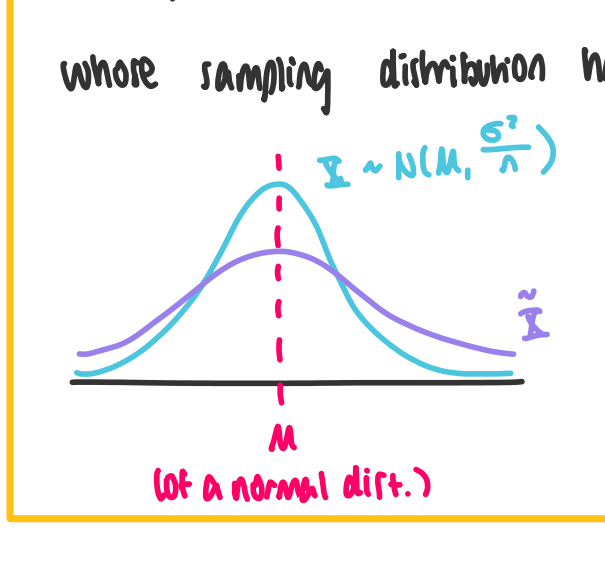
proof:
$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}[(X_i-\mu) - (\bar{X}-\mu)]^2$$
$$= \sum_{i=1}^{n}(X_i-\mu)^2 - 2[(\bar{X}-\mu)\sum_{i=1}^{n}[(X_i-\mu) + n(\bar{X}-\mu)^2]$$
$$= \sum_{i=1}^{n}(X_i-\mu)^2 - 2n(\bar{X}-\mu)(\bar{X}-\mu) + n(\bar{X}-\mu)^2$$
$$= \sum_{i=1}^{n}(X_i-\mu)^2 - n(\bar{X}-\mu)^2$$

$$E(s^2) = E\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\right]$$
$$= \frac{1}{n-1}\left[\sum_{i=1}^{n}E[(X_i-\bar{X})^2] - nE[(\bar{X}-\mu)^2]\right]$$
$$= \frac{1}{n-1}\left(\sum_{i=1}^{n}\sigma_{X_i}^2 - n\sigma_{\bar{X}}^2\right) \qquad \sigma^2_{X_i} = \sigma^2 \quad \sigma^2_{\bar{X}} = \frac{\sigma^2}{n}$$
$$= \frac{1}{n-1}\left(n\sigma^2 - n\frac{\sigma^2}{n}\right)$$
$$= \sigma^2$$



Variance of a point estimator

If $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are two unbiased estimator of θ, we could choose the estimator whose sampling distribution has the smaller variance.



Defn: If we consider all possible unbiased estimators of some parameter θ, the one with the smallest variance is the most efficient estimator of θ.

Defn: The std. dev of the sampling dist. of an estimator (or statistic) is called std. error.

$X_i \overset{iid}{\sim} N(\mu, \sigma^2)$
$\uparrow$ σ: std. dev

$\bar{X} \overset{iid}{\sim} N(\mu, \frac{\sigma^2}{n})$
$\uparrow$ $\frac{\sigma}{\sqrt{n}}$: std. error

Interval estimation

A confidence interval (CI) reports an interval of plausible values for θ.

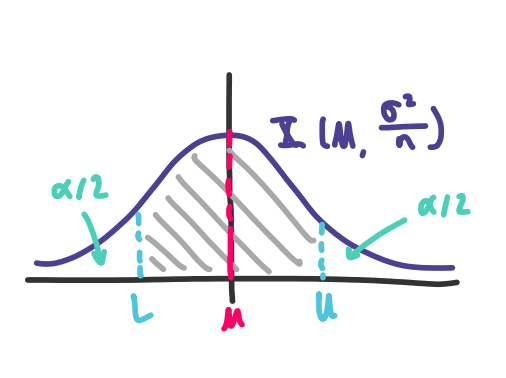
A confidence level measures the degree of reliability of the CI.

$$P(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1-\alpha, \quad 0<\alpha<1$$
res

* The interval $\hat{\theta}_L, \hat{\theta}_H$ computed from the sample is called the $100(1-\alpha)\%$ CI with a $100(1-\alpha)\%$ confidence level.

CI for μ when σ is known

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow$ if the sample is from normal or CLT



$P(L < \mu < U) = 1-\alpha$ ($\bar{x}$)

$P\left(-z_{\alpha/2} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1-\alpha$
$\hookrightarrow z \sim N(0,1)$

$$P(\underbrace{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{L} < \mu < \underbrace{\bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{U}) = 1-\alpha$$

Ex. A firm manufactures lightbulbs. If a sample of 30 has an average life of 780h, find a 95% CI for the population mean lifetime.

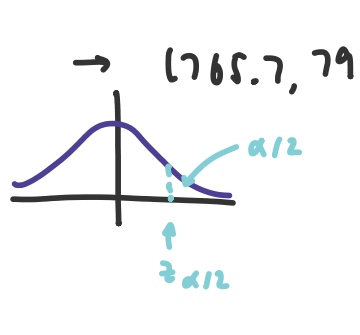
Assume $\sigma = 40h$.

$x_1, \dots, x_{20}$ } $\bar{x} = 780h$ ($\uparrow$ lifetime)

$\hat{\mu} = 780h$

$780 \mp 1.96 \frac{40}{\sqrt{50}} \rightarrow (765.7, 794.3)$

$\hookrightarrow z = \frac{0.05}{2}$



Sample size of desired error

We can be $100(1-\alpha)\%$ confident that error $|\bar{x}-\mu|$ will not exceed $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

$|\bar{x}-\mu| \le z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le e$ ← desired error limit

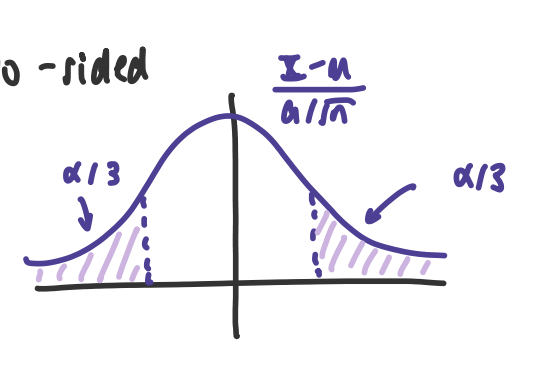
$$\boxed{n \ge \left(\frac{z_{\alpha/2}\,\sigma}{e}\right)^2}$$

Ex. for lightbulb, how large a sample is required to be 95% confident that our estimate of μ is off by less than 10h?

$$n \ge \left(\frac{\overset{z_{\alpha/2}}{1.96} \cdot \overset{\sigma}{40}}{\underset{e}{10}}\right)^2 = 61.5$$
$\hookrightarrow n = 62$ or more

CI's need not be central or two-sided

$P\left(a < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < b\right) = 1-\alpha$



one-sided CI:

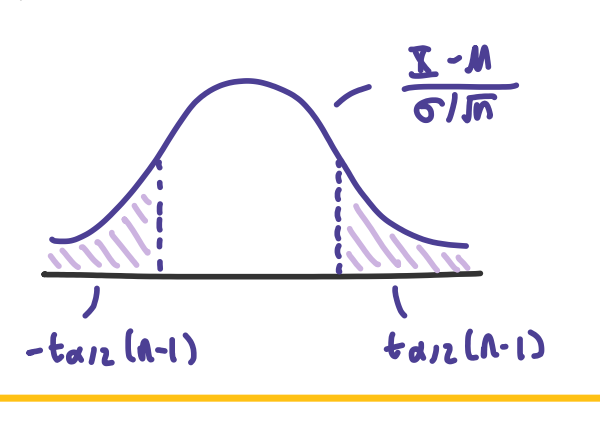
$P\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < z_\alpha\right) = 1-\alpha$

upper-one sided: $(-\infty, \bar{x} + z_\alpha\frac{\sigma}{\sqrt{n}})$

lower: $(\bar{x} - z_\alpha\frac{\sigma}{\sqrt{n}}, \infty)$

CI for μ when σ unknown:

sample from $N(\mu, \sigma^2)$



$$\bar{X} \pm t_{\alpha/2, n-1}\frac{s}{\sqrt{n}}$$

Ex. In a random sample of $n=500$ families w/ TVs.
In a particular city, 340 subscribe to HBO.
Find 95% CI for the actual proportion of families. (p)

$\hat{p} = \frac{340}{500} = \frac{\sum_{i=1}^{340} X_i}{500}$ ← Bernoulli (p)

$\bar{X} \overset{app}{\sim} N(p, \frac{pq}{n})$

$\bar{X} \sim (\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$P\left(-z_{\alpha/2} < \frac{\hat{p}-p}{\sqrt{pq/n}} < z_{\alpha/2}\right) = 1-\alpha$

$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{500}}$

$\frac{340}{500} \pm 1.96\sqrt{\frac{\frac{340}{500}\left(\frac{160}{500}\right)}{500}}$

$0.64 < p < 0.72$

prediction interval (PI)

an interval for a single new obs. X_0

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ $\quad X_0 \sim (\mu, \sigma^2)$
independent

$X_0 - \bar{X} \sim N(0, \frac{\sigma^2}{n} + \sigma^2)$

① σ known: $\bar{x} \pm z_{\alpha/2}\,\sigma\sqrt{1+\frac{1}{n}}$

② σ unknown: $\bar{x} \pm t_{\alpha/2,n-1}\,s\sqrt{1+\frac{1}{n}}$

Ex. r.s. of 20 car owners show that a car is driven on the average 23500 km/year. with $s = 3900$ km. Assume normal.

a) 95% CI for μ $\qquad \alpha = 1-0.95 = 0.05$

$\bar{x} = 23500, \quad s = 3900 \qquad t_{\alpha/2, n-1}$

$\hat{\mu} = 23500$

$23500 \pm \underset{t_{0.025,19}}{2.093}\frac{3900}{\sqrt{20}}$

$21674 < \mu < 25326$

b) If I select a new sample, 95% PI?

$\hat{x}_0 = 23500$

$23500 \pm 2.093 \cdot 3900\sqrt{1+\frac{1}{20}}$

$15135 < x_0 < 31865$

Two samples from normal

① variances known

$\bar{X}_1 - \bar{X}_2 \sim N(\mu_1-\mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$

$$P\left(-z_{\alpha/2} < \frac{(\bar{X}_1-\bar{X}_2) - (\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} < z_{\alpha/2}\right) = 1-\alpha$$

$(1-\alpha)100\%$ CI: for $\mu_1-\mu_2$: $\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$

② variances unknown but equal

$\sigma_1^2 = \sigma_2^2 = \sigma^2$

$\bar{X}_1 - \bar{X}_2 \sim N(\mu_1-\mu_2, \sigma^2(\frac{1}{n_1}+\frac{1}{n_2}))$

$$\hat{\sigma}^2 = s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$
$\uparrow$ pooled var.

$(1-\alpha)100\%$ CI: $\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2, n_1+n_2-2}\, s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$ for $\mu_1-\mu_2$

③ variances unknown & unequal

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1-\mu_2)}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} \sim t(\nu)$$

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$
← round down to nearest integer

$(1-\alpha)100\%$ CI for $\mu_1-\mu_2$: $\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2,\nu}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$

Ex. A cab company wants to choose between brand A & B tires.
An exp. w/ 12 from each brand are conducted. Tires run until they wear out.
A: $\bar{x}_A = 36700$ km, B: $\bar{x}_B = 38100$ km.
Compute 95% CI for $\mu_A - \mu_B$.

(a) assume $\sigma_A = 5000$ km, $\sigma_B = 6100$ km (case ①)

$\bar{x}_A - \bar{x}_B \pm z_{\alpha/2}\sqrt{\frac{\sigma_A^2}{n_A}+\frac{\sigma_B^2}{n_B}} \Rightarrow (-6263, 2663)$ ($\alpha/2$: 0.025)

(b) assume $s_A = 5000$ km, $s_B = 6000$ km, and $\sigma_A^2 = \sigma_B^2$ (case ②)
(-6522, 2922)

(c) assume $s_A = 5000$ km, $s_B = 6000$ km, and $\sigma_A^2 \ne \sigma_B^2$ (case ③)
(-6536, 2936)

Decision fail to reject $\mu_A = \mu_B$

Paired observations

Two observations on same individual

$D_i = X_{1i} - X_{2i} \qquad i = 1, \dots, n$

$var(D_i) = var(X_{1i} - X_{2i}) = \sigma_1^2 + \sigma_2^2 - \underbrace{2cov(X_{1i}, X_{2i})}_{\text{generally positive}}$

95% CI for $\mu_D = \mu_1 - \mu_2$:

$\bar{d} \pm t_{\alpha/2, n-1}\frac{s_d}{\sqrt{n}}$ ← n pairs for 2n observations

Ex. College offers English courses
20 students are given TOEFL exams before & after.

student	pre-course	post-course	d_i (post-pre)
1	30	29	-1
2	28	30	2
3	31	32	1
⋮	⋮	⋮	⋮
20	29	32	3

$\bar{d} = \frac{\sum_{i=1}^{20} d_i}{20} = 1.45$

$s_d = \sqrt{\frac{\sum_{i=1}^{20}(d_i - \bar{d})^2}{20-1}} = 3.203$

95% CI for $\mu_{post} - \mu_{pre}$: $1.45 \pm \underset{t_{0.025,19}}{2.02}\frac{3.203}{\sqrt{20}} \Rightarrow (-0.049, 2.949)$

$X_1, \dots X_n \sim N(\mu, \sigma^2)$

$\frac{(n-1)}{\sigma^2}s^2 \sim \chi^2(n-1)$

$P\left(\chi^2_{1-\alpha/2, n-1} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2, n-1}\right) = 1-\alpha$

95% CI for σ^2: $\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$

Ex. Tire σ_A^2 for $n_A = 12$, $s_A^2 = 5000$ km
CI: (2505, 1441.3) $\quad \hat{\sigma}_A^2 = 5000$
Two samples:

$\frac{\sigma_2^2 s_1^2}{\sigma_1^2 s_2^2} \sim F(\overset{\nu_1}{n_1-1}, \overset{\nu_2}{n_2-1})$

$\frac{\sigma_1^2}{\sigma_2^2}$ CI: $\frac{s_1^2}{s_2^2}\frac{1}{f_{\alpha/2,\nu_1,\nu_2}}, \quad \frac{s_1^2}{s_2^2}f_{\alpha/2,\nu_2,\nu_1}$

Ex. Tire example

$\sigma_A^2 = \sigma_B^2$

$\frac{\hat{\sigma}_A^2}{\hat{\sigma}_B^2}$ ← 95% CI

$\frac{5000^2}{6100^2}\frac{1}{3.474}, \quad \frac{5000^2}{6100^2}\ 3.474$

fail to reject $\sigma_A^2 = \sigma_B^2 \qquad (0.193, 2.33)$

Maximum Likelihood Estimation (MLE)

Given a r.s. from a dist. w/ parameter θ, the joint dist. is

$f(x_1, x_2, \dots x_n|\theta) = f(x_1|\theta)f(x_2|\theta)\dots f(x_n|\theta)$

Define likelihood function

$L(\theta|x_1, \dots x_n) = f(x_1, \dots x_n|\theta)$

MLE, $\hat{\theta}$, is the value of θ that maximizes the likelihood function.

Ex. 3 items from an assembly line are inspected: 2N followed by D.
Estimate p, proportion of nondefective in process.

$L(p|\underbrace{x_1, x_2, x_3}_{data}) = p \cdot p(1-p) = p^2 - p^3$

$\frac{\partial(p^2-p^3)}{\partial p} = 2p - 3p^2 = 0$

$\hat{p} = \frac{2}{3}$ ← MLE

Ex. Random sample from $N(\mu, \sigma^2)$ find MLE of μ, σ^2

$L(x_1, \dots x_n|\mu, \sigma^2) = \frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}}\exp\left[-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{2}\right)^2\right]$

$\ln L = -\frac{n}{2}\ln(2\pi) = -\frac{n}{2}\ln\sigma^2 - \frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{2}\right)^2$

$\frac{\partial \ln L}{\partial \mu} = \sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right) = 0, \quad \hat{\mu} = \bar{x}$

$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} = \sum_{i=1}^{n}(x_i-\mu)^2 = 0, \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$

check 2nd derivatives < 0, so maximum

Chapter 10: One and Two Sample Tests of Hypotheses

Friday, March 20, 2026 3:23 PM

Reality

Decision

	H_0 true	H_0 false
fail to reject H_0	correct decision ☺	Type II → miss error
reject H_0	Type I error (false alarm)	☺ hit

H_0: defendant is innocent

H_1: defendant is guilty

↳ H_0 maintained unless H_a supported "beyond a maintainable doubt"

Defn: A statistical hypothesis is an assertion concerning one or more populations.

H_0 null hypothesis: what a complete skeptic would say or "status quo"

H_1 (H_a) alternative hypothesis: what we "want" to show

Ex. Company tests whether mean weight exceeds 10 pounds. r.s. of 49 samples examined.

$\bar{x} = 11$ lbs Assume $\sigma = 2.8$ ← population => use z

$H_0: \mu \le 10$

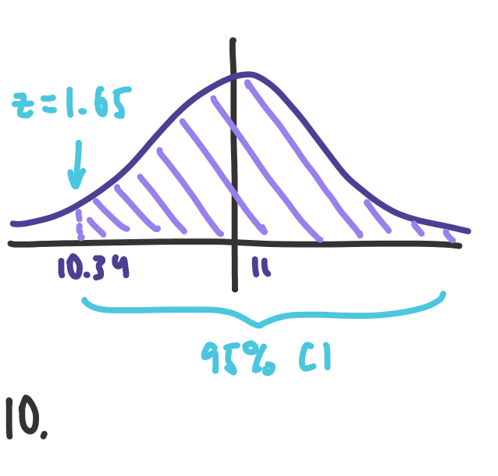
$H_a: \mu > 10$

Method 1: find 95% CI, check if 10 is in

$\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$ lower bound

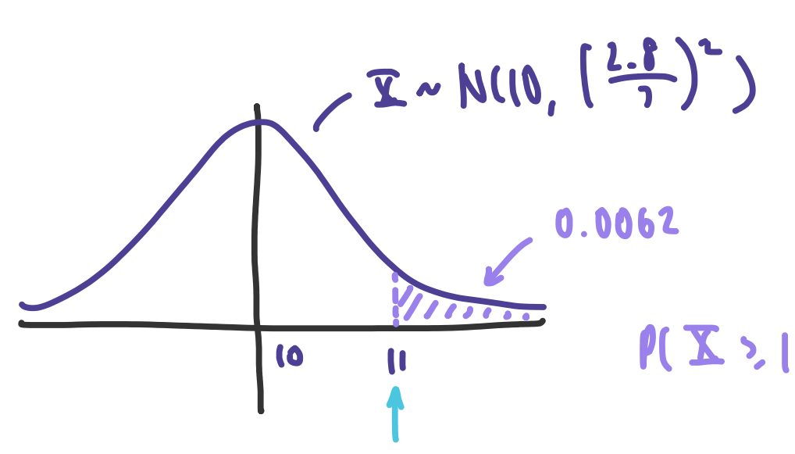
$11 - 1.65 \frac{2.8}{\sqrt{49}} \Rightarrow (10.34, \infty)$

↑ $\hat{\mu}$



Reject H_0 because CI does not include 10.

Method 2: Assume H_0 true, $\mu = 10$, check if $\bar{x} = 11$ has p-value less than 0.05



$z = 2.5$ "test statistic value"

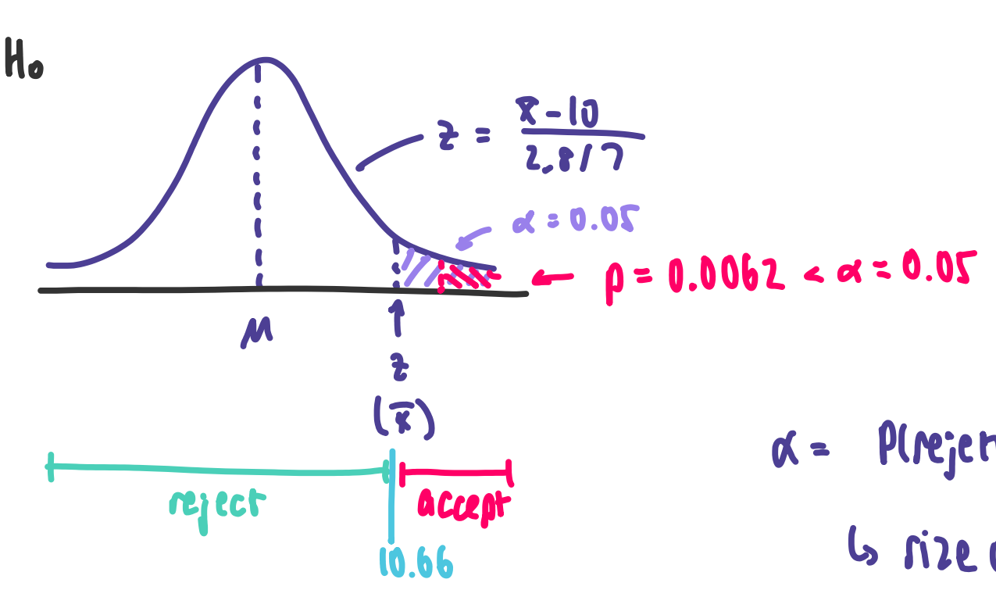
$z = \frac{11-10}{\frac{2.8}{7}} = 2.5$

$P(\bar{X} \ge 11 \mid H_0 \text{ true}) = P(z > 2.5)$

$= 1 - 0.9938$

$= \underbrace{0.0062}_{\text{p-value}} < 0.05 \Rightarrow$ reject H_0

under H_0



α = P(rejecting H_0 | H_0 true)

↳ size of a test significance value

$z_{central} = 1.65$

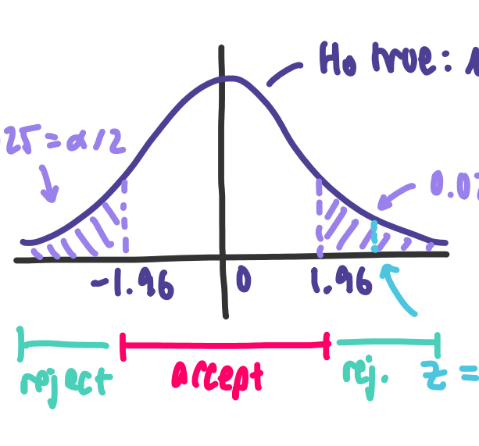
$\bar{X}_{cr} = 10 + 1.65 \frac{2.8}{7} = 10.66$

Ex. Assume we are interested in testing.

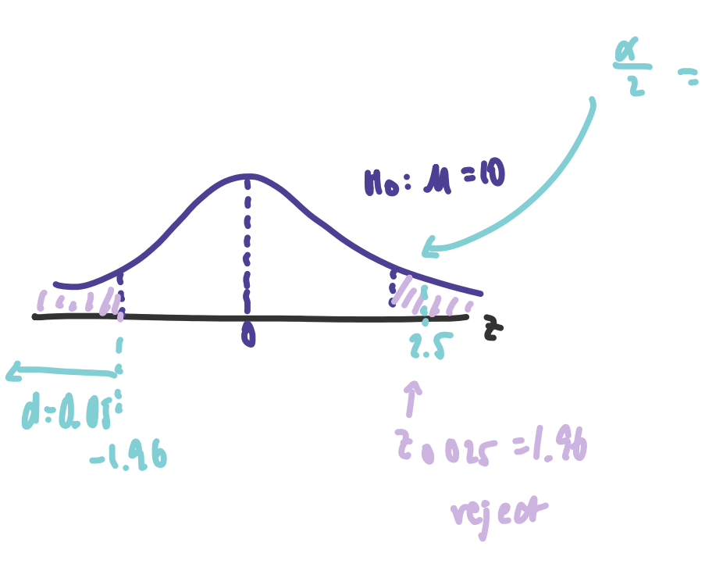
$H_0: \mu = 10$

$H_1: \mu \ne 10$

$\bar{X} \sim N(10, (\frac{2.8}{7})^2)$



$z = \frac{11-10}{2.8/7} = 2.5 \Rightarrow$ reject



Type I Error (α): P(reject H_0 | H_0 true) risk in incorrectly rejecting H_0

$P(z > 2.5) + P(z < -2.5) = 0.062 \times 2 = 0.0124 < 0.05$

95% CI for μ two sided

$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ $11 \pm 1.96 \frac{2.8}{7} \to (10.216, 11.784)$

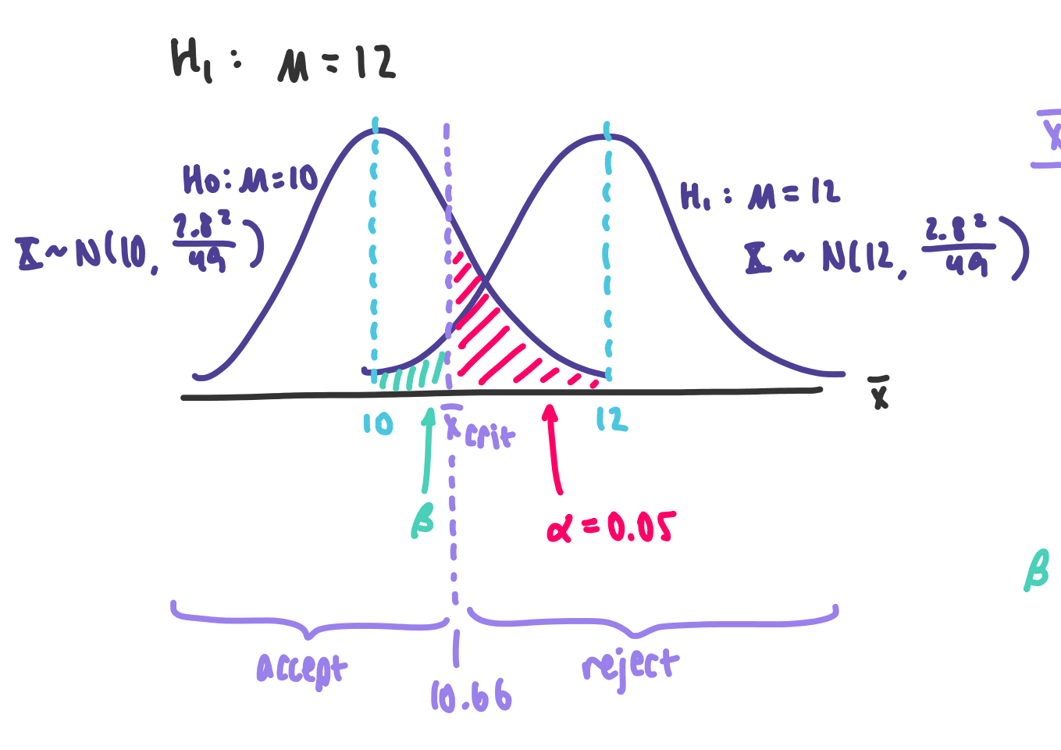
excludes 10 so reject

Type II error (β): P(not reject H_0 | H_0 is false)

$= 1 - \underbrace{P(\text{reject } H_0 \mid H_0 \text{ is false})}_{\text{power (say guilty when guilty)}}$

$H_0: \mu = 10$

$H_1: \mu = 12$

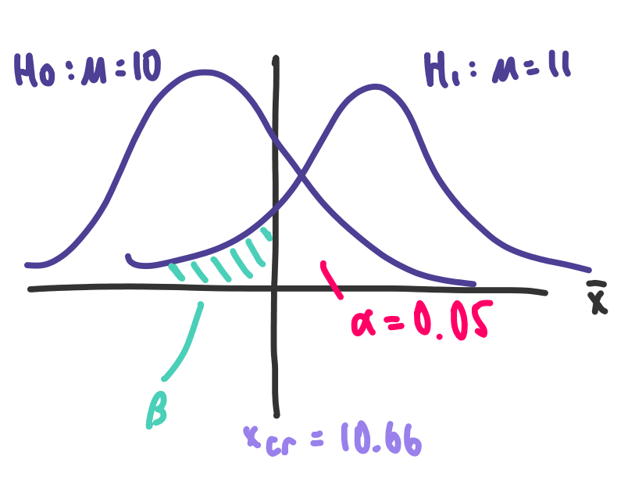


$\frac{\bar{x}_{cr} - 10}{2.8/7} = 1.65$ ← z_{cr}

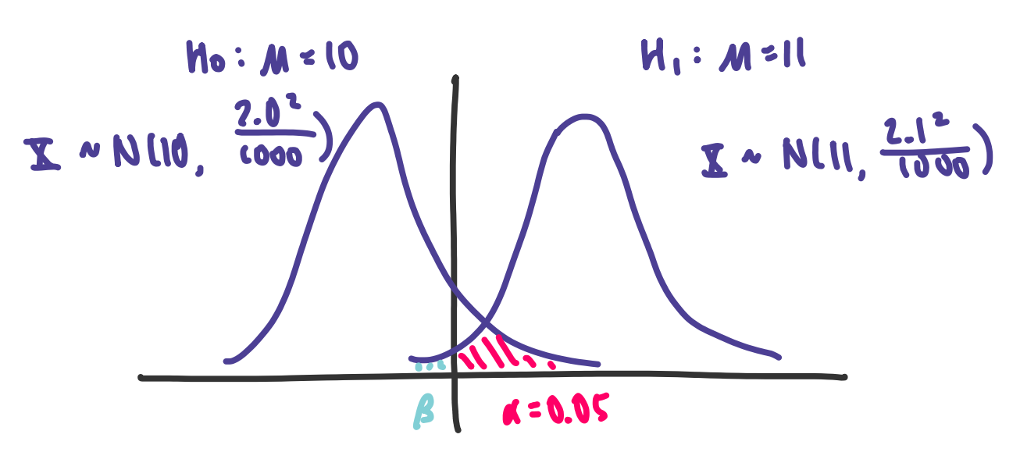
$\beta = 1 - P(z < \frac{10.66-12}{2.8/7})$

$= P(z < -3.35) = 0.0004$

What if $H_1: \mu = 11$



$\beta = P(z < \frac{10.66-11}{\frac{2.8}{7}}) = P(z < -0.85) = 0.20$



$\frac{\bar{x}_{cr} - 10}{\frac{2.8}{\sqrt{1000}}} = 1.65$

$\bar{x}_{cr} = 10.15$

$\beta = P(z < \frac{10.15 - 11}{\frac{2.8}{\sqrt{1000}}})$

$= P(z < -9.6)$

≈ 0

Ex. freight train again but assume normal

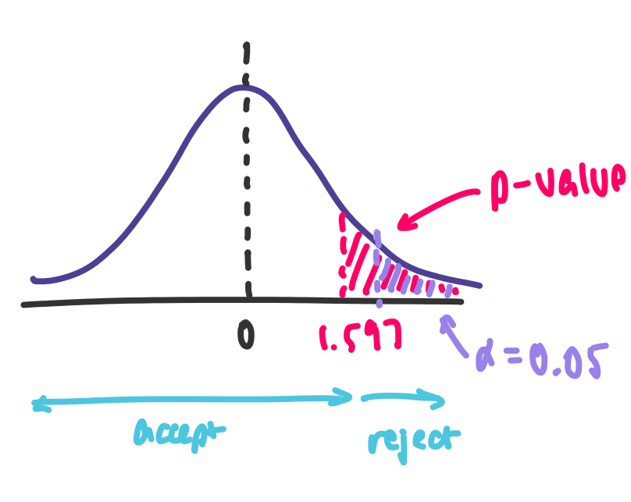
r.s. of 20 orders, $\bar{x} = 11$, $s = 2.8$ lbs ← sample mean/std => use t

$H_0: \mu = 10$

$H_1: \mu > 10$

Recall $T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$

$H_0: \frac{\bar{X} - 10}{\frac{2.8}{\sqrt{20}}} \sim t(19)$



t-value $= \frac{11-10}{2.8/\sqrt{20}} = 1.597$

p-value $= 0.0634 > 0.05 = \alpha$

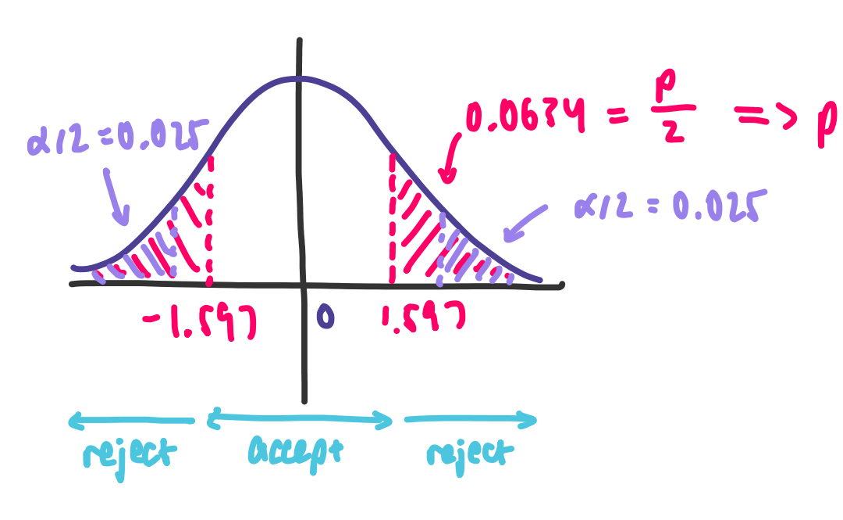
fail to reject H_0

Ex. same as previous, but now:

$H_0: \mu = 10$

$H_1: \mu \ne 10$

$t = \frac{11-10}{2.8/\sqrt{20}} = 1.597$



$0.0634 = \frac{p}{2} \Rightarrow p = 0.1268 > 0.025 = \alpha$

fail to reject H_0

Two samples

Two ind. samples of size n_1 and n_2 from 2 populations w/ $\mu_1, \sigma_1^2, \mu_2, \sigma_2^2$ parameters.

$H_0: \mu_1 - \mu_2 = d_0$ (constant)

$H_1: \mu_1 - \mu_2 \ne d_0$

① known var. σ_1^2, σ_2^2

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

② unknown but equal var.

population normal

pooled t-test

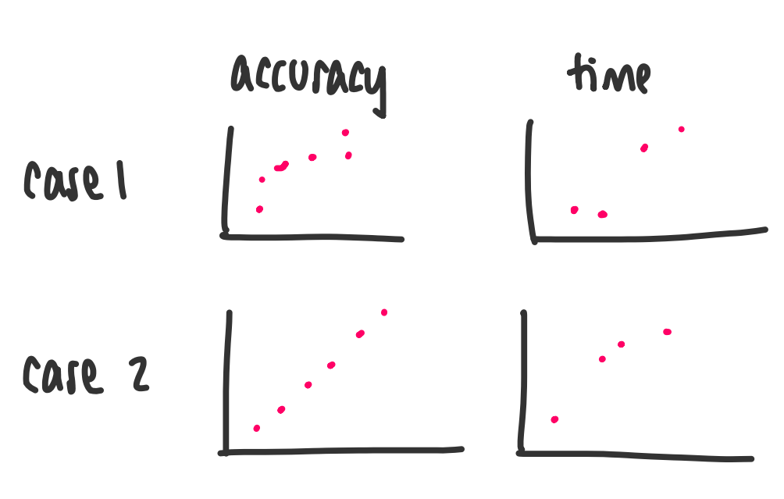
$$\underset{\uparrow \text{ pooled var.}}{S_p^2} = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1+n_2-2)$$

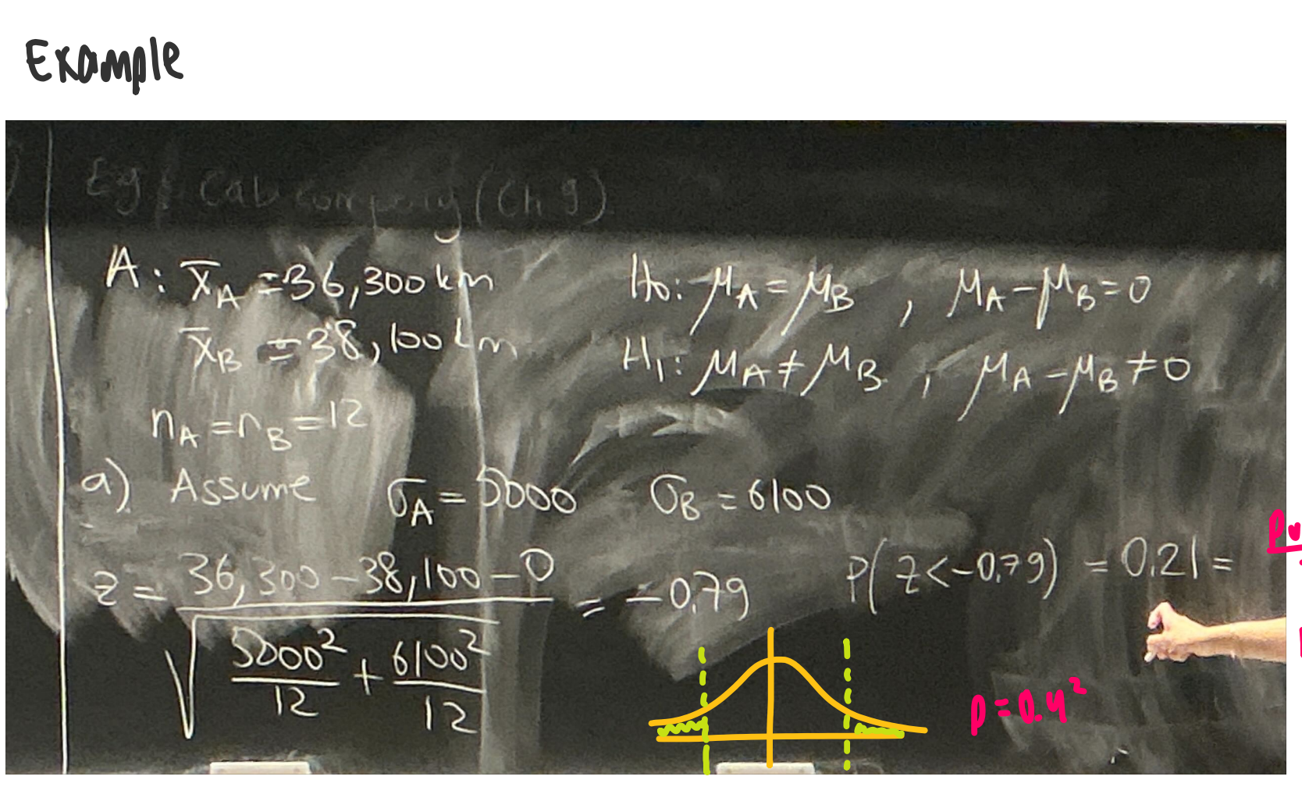
③ unknown and unequal var.

normal dist.

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \overset{\sim}{_{app}} t(\nu) \text{ ← see ch.9}$$



Example



Eg: Cab company (Ch 9)

A: $\bar{X}_A = 36{,}300$ km

$\bar{X}_B = 38{,}100$ km

$n_A = n_B = 12$

a) Assume $\sigma_A = 5000$ $\sigma_B = 6100$

$$z = \frac{36{,}300 - 38{,}100 - 0}{\sqrt{\frac{5000^2}{12} + \frac{6100^2}{12}}} = -0.79$$

$H_0: \mu_A = \mu_B$, $\mu_A - \mu_B = 0$

$H_1: \mu_A \ne \mu_B$, $\mu_A - \mu_B \ne 0$

$P(z < -0.79) = 0.21 = \frac{\text{pval}}{2}$

$p = 0.42$

$p = 0.4$

b) assume $s_A = 5000$

$s_B = 6100$ $\sigma_A^2 = \sigma_B^2$

$t = \frac{-1800}{s_p\sqrt{\frac{1}{12}+\frac{1}{12}}} = -0.791$ ↳ $t(22)$

s_p ↑ 5577

$P(T < -0.791) = 0.219$

p-val $= 0.438$

c) assume $s_A = 4000$, $s_B = 6100$

$\sigma_A^2 \ne \sigma_B^2$

$t = -0.79$

$\nu = 21$

$p = 0.4384$

Goodness of fit test

Ex. A die is cast 120 times

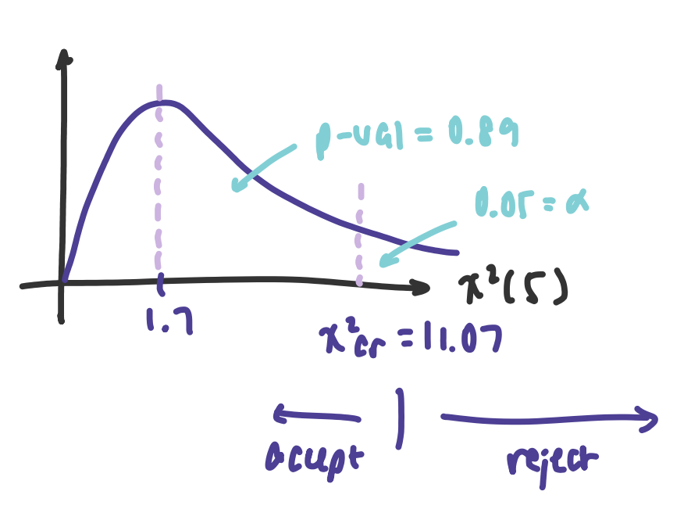
face	1	2	3	4	5	6
observed	20	22	17	18	19	24
expected if dice fair	20	20	20	20	20	20

$$\chi^2 = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i} \overset{app}{\sim} \chi^2(k-1)$$ ← smaller χ^2 values indicate good fit

for approximation, each $e_i \ge 5$

Test statistic

χ^2 $\frac{(20-20)^2}{20} + \frac{(22-20)^2}{20} + \dots + \frac{(24-20)^2}{20} = 1.7$



Ex. car battery life in years

$n = 40 \Rightarrow 2.2, 4.1, 3.5, \dots, 4.2$ } $N(3.5, 0.7^2)$ ← $\hat{\mu}$, $\hat{\sigma}$

class boundaries	o_i		e_i	
1.45 - 1.95	2	7	0.5	8.5
1.95 - 2.45	1		2.1	
2.45 - 2.95	4		5.9	
2.95 - 3.45	15		10.3	
3.45 - 3.95	10		10.7	
3.95 - 4.45	5		7.0	10.5
4.45 - 4.95	3		3.5	

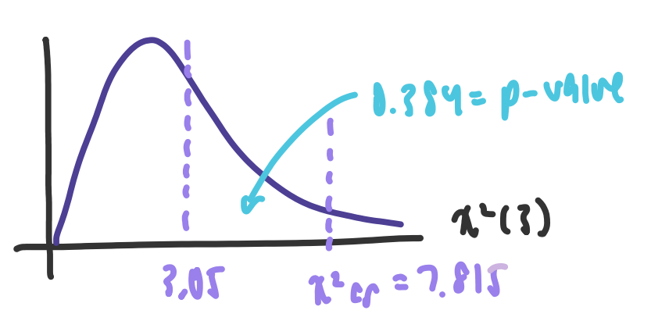
$P(2.95 < \bar{X} < 3.45 \mid N(3.5, 0.7^2)) = P\left(\frac{2.95-3.5}{0.7} < z < \frac{3.45-3.5}{0.7}\right)$

$= 0.2573$

$e_4 = 0.2573 \times 40 = 10.3$

$\chi^2 = \frac{(7-8.5)^2}{8.5} + \frac{(15-10.3)^2}{10.3} + \frac{(10-10.7)^2}{10.7} + \frac{(8-10.5)^2}{10.5}$

$= 3.05$



Chapter 11: Simple linear regression

Friday, March 27, 2026 3:31 PM

Studies the relation between a quantitative response variable x, and a single explanatory variable Y

$$Y = \underbrace{\beta_0}_{\text{const.}} + \underbrace{\beta_1}_{\text{const.}} x + \varepsilon \qquad E(\varepsilon) = 0, \quad var(\varepsilon) = \sigma^2$$

Y: response var, dep. var

x: explanatory var, predictor var, independent var

Y: price of house

x: square footage

x_2: age of a house

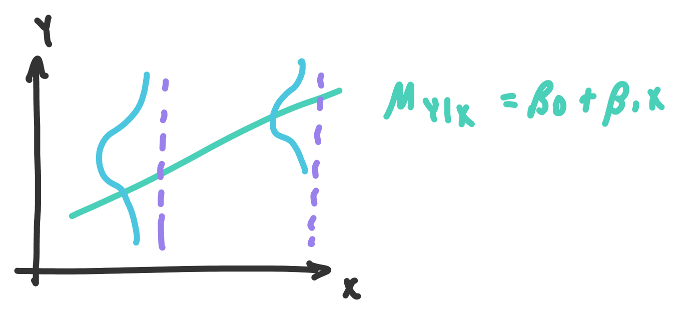
multiple linear regression: $Y = \beta_0 + \beta_1 x + \beta_2 x_2 + \varepsilon$

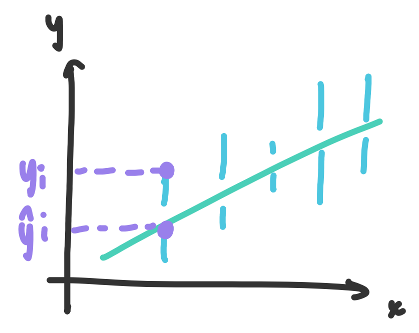
$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

linear in coefficients (β, not x's)

$E(Y) = \beta_0 + \beta_1 x$

$var(Y) = \sigma^2$





Least squares and the fitted value

$\hat{y} = b_0 + b_1 x \qquad b_0 : \hat{\beta}_0 \qquad b_1 : \hat{\beta}_1$

Defn: Residual

Given a set of regression data (x_i, y_i), $i = 1, \dots n$ and a fitted model $\hat{y}_i = b_0 + b_1 x_i$, the i^{th} residual is $e_i = y_i - \hat{y}_i$, $i = 1, \dots n$

The least squares method estimates β_0, β_1 by minimizing

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

SSE: sum of squares of error

$$\frac{\partial SSE}{\partial b_0} = -2 \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) = 0$$

$$\frac{\partial SSE}{\partial b_1} = 0$$

$$\hat{\beta}_1 = b_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad \begin{matrix} \leftarrow S_{xy} \\ \leftarrow S_{xx} \end{matrix}$$

$$\hat{\beta}_0 = b_0 = \bar{y} - b_1 \bar{x}$$

$$\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2), \quad i = 1, \dots, n$$

$$y_i \overset{ind}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

β_0, β_1 : true coefficients

b_0, b_1 : estimates

B_0, B_1 : estimators

$$B_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

properties of least squares estimators (LSE)

$$B_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{\sum_{i=1}^{n} (x_i - \bar{x}) y_i}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$E(B_1) = \beta_1$

$B_1 \sim N(\beta_1, \sigma^2 / S_{xx})$

$$B_0 \sim N\left(\beta_0, \frac{\sum_{i=1}^{n} x_i^2}{n S_{xx}} \sigma^2\right)$$

$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$

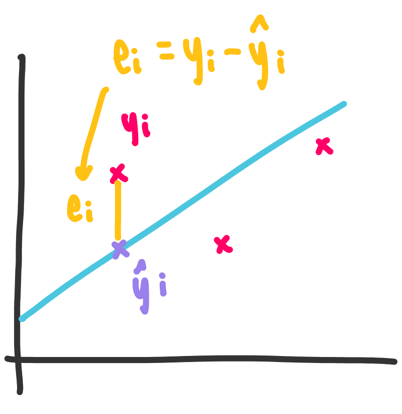
$S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2$

$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$

mean-squared error:

$$\hat{\sigma}^2 = s^2 = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-2} = \frac{SSE}{n-2}$$

$E(s^2) = \sigma^2$



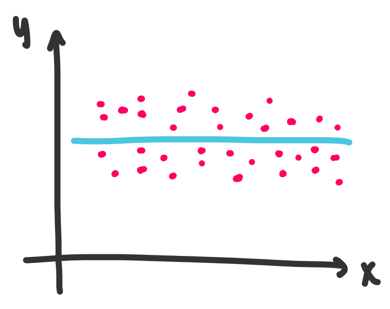
$H_0 : B_1 = 0$

$H_1 : B_1 \neq 0$

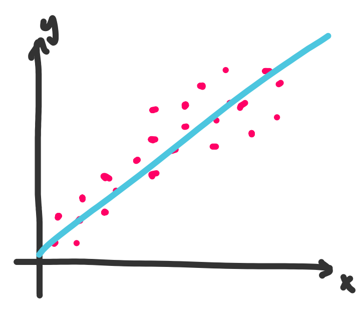
under null $\quad T = \dfrac{B_1 - \beta_1}{S / \sqrt{S_{xx}}} \sim t(n-2)$ — here 0

$100(1-\alpha)\%$ CI for B_1 : $\quad b_1 \pm t_{\alpha/2, n-2} \underbrace{\frac{s}{\sqrt{S_{xx}}}}_{\text{se. of estimate}}$

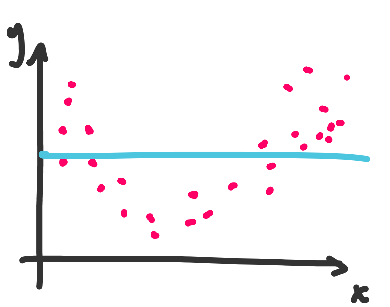
2-sided



fail to reject



reject



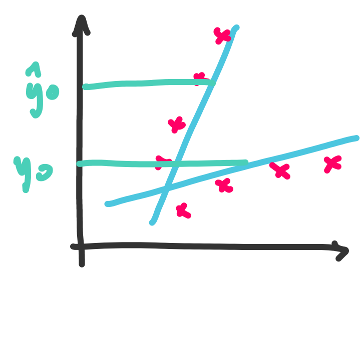
fail to reject

$y_i = \beta_0 + \beta_0 x_i + \beta_0 x_i^2 + c_i$

Prediction

The eqn $\hat{y} = b_0 + b_1 x$ can be used to estimate the mean response $M_{y|x_0}$ at $x = x_0$

$$\hat{y}_0 \sim N\left(\beta_0 + \beta_1 x, \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]\right)$$



$100(1-\alpha)\%$ CI, $\hat{y}_0 \pm t_{\alpha/2, n-2} \underbrace{s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}_{\text{s.e. estimate}}$

two-sided

Predicting a new observation Y_0

$$\hat{y}_0 \pm t_{\alpha/2, n-2} \, s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$